

The Compass



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Community Mathematics Centre

Schools of the Krishnamurti Foundation India
(KFI)

Statement of intent

The magazine is published by the Community Mathematics Centre. It is a collaborative initiative of the schools of the Krishnamurti Foundation India (KFI).

Its intention is:

- To provide material for mathematics and science teachers who teach at the high school or middle school levels, and for students who study at these levels.
- To serve as a platform where teachers and students can share their findings and discoveries with others who share their love for the subject.

In addition to mathematical content, in each issue we shall share a few ‘snippets’ from the world of science — short extracts abstracted from articles in online magazines, selected on the basis of intrinsic interest.

The Compass is published as an online magazine free of cost. Articles may be downloaded and used freely by teachers, but with proper attribution.

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Colour coding

The magazine has articles addressed to teachers who teach at different levels: Lower Primary level; Upper Primary and High School levels; and Senior Secondary and Junior College levels. There is also a Problem Corner. In addition we have the Snippets.

For the convenience of the reader, we use the following colour coding to distinguish between the various levels. Of course, such a division can never be exact. So, our apologies in advance for errors of judgement!

- Primary & Upper Primary (Classes 4, 5, 6, 7)
- Upper Primary & High School (Classes 6, 7, 8, 9, 10)
- Senior Secondary & Junior College (Classes 10, 11, 12)
- Problem Corner
- Snippets

Fast Calculations – 2

We continue our study of computational shortcuts (or, “the art of doing fast calculations and impressing an audience”). In this issue we explain the logic behind the shortcut given in the July 2024 issue for computing squares (of some numbers). In that article we looked at a number of examples of the following kind.

To find the square of 12

- $12 + 2 = 14$
- $2^2 = 4$
- Hence $12^2 = 144$

This particular shortcut only works when the two-digit number has tens digit 1. So the number must be of the form $10 + a$ for some digit a .

For example, if the number is 16, we write it as $10 + 6$, with $a = 6$.

Squaring this, we get

$$(10 + a)^2 = 100 + 20a + a^2.$$

This may be written as

$$10(10 + a + a) + a^2.$$

Let's see what each part of this expression means.

- (A) $10 + a$ is the given number, and $10 + a + a$ means that we **add** the units digit a to the given number.
- (B) $10(10 + a + a)$ means that we **multiply** the result from (A) by 10. (This is the same as inserting a 0 at the units end of that number.)
- (C) $10(10 + a + a) + a^2$ means that we **add** the square of the unit's digit a to the result from (B).
- (D) If a^2 happens to be a two-digit number, then we must "carry" its tens digit to the tens digit of $10(10 + a + a)$, just as we do in addition.

From the explanation, we see why the shortcut works only for two-digit numbers with tens digit 1.

But we also see a way forward! Let's see what will happen if the tens digit is not 1.

Suppose the tens digit is 2. If the units digit is a , the number is $20 + a$. Squaring this, we get

$$(20 + a)^2 = 400 + 40a + a^2 = 20(20 + a + a) + a^2.$$

The form of this leads us to the 'shortcut' with steps as follows:

- $20 + a$ is the given number, and $20 + a + a$ means that we **add** the units digit a to the given number.
- $20(20 + a + a)$ means that we **multiply** the result from (A) by 20. (This is the same as multiplying by 2 and inserting a 0 at the units end.)

- To this we add a^2 , taking care of the carry-over as needed.

Now let's try it out. It does work!

To find the square of 23

- $23 + 3 = 26$
- $20 \times 26 = 520$ (we can also write this as $52_$)
- $3^2 = 9$
- Hence $23^2 = 520 + 9 = 529$

To find the square of 27

- $27 + 7 = 34$
- $20 \times 34 = 680$ (we can also write this as $68_$)
- $7^2 = 49$
- $68 + 4 = 72$; or we can directly do $680 + 49$
- Hence $27^2 = 729$

Try this shortcut with a few other two-digit numbers starting with 2.

Exercises for you

We leave you with these interesting challenges.

1. Find a shortcut for working out the squares of two-digit numbers whose tens digit is 3.
2. Find a shortcut for working out the squares of two-digit numbers whose tens digit is 4.

Pretty Number Patterns – 2

We are going to continue to explore pretty patterns in this issue. Let us start by explaining the “mischievous fractions” phenomenon. Recall that these are fractions where we can ‘illegally’ cancel digits from the numerator and denominator of a fraction, and no error seems to result; e.g.,

$$\frac{1\cancel{6}}{\cancel{6}4} = \frac{1}{4}.$$

To understand this we must look beneath the surface — which means we have to bring in algebra. Let the equality be

$$\frac{10a + b}{10b + c} = \frac{a}{c}.$$

Here we have ‘cancelled’ the common digit b from the numerator and the denominator. For which values of a, b, c does such an equality hold?

Cross-multiplying, we get the following equality:

$$10ac + bc = 10ab + ac.$$

This yields:

$$9ac = b(10a - c)$$

$$\therefore b = \frac{9ac}{10a - c}.$$

So for the “illegal equality” to hold, a, b, c must satisfy the above relation.

Let’s see if this works for the equality we started with, $16/64 = 1/4$. We have here, $a = 1, b = 6, c = 4$. So:

$$\frac{9ac}{10a - c} = \frac{9 \times 1 \times 4}{10 \times 1 - 4} = \frac{36}{6} = 6,$$

and since $b = 6$, everything works out the way it should.

Are there any more such possibilities? There are now two ways of proceeding. Let’s keep a fixed at 1 and look for a value of c for which $10a - c$ is a divisor of $9ac$. Since $a = 1$, this means that $10 - c$ is a divisor of $9c$.

Let’s check each value of c from 1 to 9 and identify the values that work. We prepare a table and see what emerges.

c	1	2	3	4	5	6	7	8	9
$9c$	9	18	27	36	45	54	63	72	81
$10 - c$	9	8	7	6	5	4	3	2	1
b	1			6	9				

In the last row, we have filled in only those entries where the division yields a digit. (If $c = 7$, we get $b = 63/3 = 21$ which is not a digit. The same thing happens when $c = 8$ or 9 . So these entries have not been filled.)

We now have the following possibilities:

- $a = 1, b = 1, c = 1$;
- $a = 1, b = 6, c = 4$; and
- $a = 1, b = 9, c = 5$.

The first solution does not yield anything interesting; we get

$$\frac{1\cancel{1}}{\cancel{1}1} = \frac{1}{1}.$$

Hmm. Not at all interesting! We discard this.

The other two possibilities are of interest:

$$\frac{1\cancel{6}}{\cancel{6}4} = \frac{1}{4}$$

and

$$\frac{1\cancel{9}}{\cancel{9}5} = \frac{1}{5}.$$

The first one is the one we started out with; but the second one is new — and we have obtained it purely through our analysis! We see here the great power of algebra.

We're not done yet, of course; we must try out $a = 2, a = 3, \dots$. But we shall leave this task for you. Please find out whether there are any more mischievous fractions! At the end of your analysis you should be sure that you have found *all* possible instances of illegal cancellation involving fractions in which the numerator and denominator are two-digit numbers.

After you are done, you could also explore illegal cancellations of this kind:

$$\frac{166}{664} = \frac{1}{4}, \quad \text{i.e.,} \quad \frac{1\cancel{6}\cancel{6}}{\cancel{6}\cancel{6}4} = \frac{1}{4}.$$

Find out if there are more instances of this kind. (There are!)

Number sequences

Number sequences are an extremely fertile area of mathematics; they offer rich possibilities for exploration and deep study.

There are sequences of all kinds. For example:

- The sequence of natural numbers: $1, 2, 3, 4, 5, \dots$;
- The sequence of odd numbers: $1, 3, 5, 7, 9, \dots$;
- The sequence of perfect squares: $1, 4, 9, 16, 25, \dots$;
- The sequence of doubling numbers: $1, 2, 4, 8, 16, \dots$;
- The sequence of prime numbers: $2, 3, 5, 7, 11, \dots$

Each of these sequences is rich in properties.

The simplest sequence is surely the *constant sequence* in which all the numbers are the same — but this is too simple to be of interest!

Next in line is the *arithmetic sequence* in which the differences between pairs of consecutive numbers are the same. For example, the first two sequences shown above. We meet such sequences frequently. In this article, we shall study only arithmetic sequences. As their definition is so simple, it may seem that there is not much to talk about; but this is certainly not the case, as we shall now see.

The sequence of natural numbers — a closer look

Let's now study the sequence of natural numbers

$$1, 2, 3, 4, 5, 6, 7, 8, \dots$$

What interesting properties do we find? Here are a few. What is interesting is that *these properties hold for any and every arithmetic sequence*.

Additive properties

Let's start with properties that mainly involve addition (and subtraction).

- Take any three consecutive numbers a, b, c in the sequence. You will find that the sum of the first and third numbers is twice the middle number. That is, $a + c = 2b$.

For example, take the triples $(7, 8, 9)$ and $(13, 14, 15)$. In the first one, $7 + 9 = 16 = 2 \times 8$, and in the second one, $13 + 15 = 28 = 2 \times 14$. Work out for yourself why this will always be true.

- Take any three consecutive numbers a, b, c in the sequence. You will find that their sum is always a multiple of 3.

For example, take the triples $(7, 8, 9)$ and $(13, 14, 15)$. In the first one, $7 + 8 + 9 = 24 = 3 \times 8$, and in the second one, $13 + 14 + 15 = 42 = 3 \times 14$. Work out for yourself why this will always be true.

- Take any four consecutive numbers a, b, c, d in the sequence. You will find that the sum of the first and last numbers is equal to the sum of the middle two numbers. That is, $a + d = b + c$.

For example, take the quadruples $(7, 8, 9, 10)$ and $(23, 24, 25, 26)$. We see that $7 + 10 = 8 + 9$ and $23 + 26 = 24 + 25$. Work out for yourself why this will always be true.

- Take any four consecutive numbers a, b, c, d in the sequence. You will find that their sum is *never* a multiple of 4.

For example, take the quadruples $(7, 8, 9, 10)$ and $(23, 24, 25, 26)$. We see that $7 + 8 + 9 + 10 = 34$ and $23 + 24 + 25 + 26 = 98$ are not multiples of 4. Work out for yourself why this will always be true.

- Take any five consecutive numbers a, b, c, d, e in the sequence. What interesting additive properties can you spot that are *always* true?
- Take any six consecutive numbers a, b, c, d, e, f in the sequence. What interesting additive properties can you spot that are *always* true?

In the last two cases we shall leave you with the task of searching for more such properties and also working out *why* they are always true. (Yes, the ‘why’ matters!)

Multiplicative properties

Arithmetic sequences also have a number of multiplicative properties. Some of these are both pretty and surprising. Here are just two of many.

- Take any three consecutive numbers a, b, c in the sequence. Multiply the two outer numbers, and square the middle number. You will find that the second quantity always exceeds the first by 1. That is, $b^2 - ac = 1$. This is invariably true.

For example, take the triples $(7, 8, 9)$ and $(9, 10, 11)$. In the first one, $8^2 = 64$ and $7 \times 9 = 63$. In the second one, $10^2 = 100$ and $9 \times 11 = 99$. In both cases we see a difference of 1, as expected.

- Take any four consecutive numbers a, b, c, d in the sequence. Multiply the two outer numbers, and then the two inner numbers. You will find that the second quantity always exceeds the first by 2. That is, $bc - ad = 2$. This is invariably true.

For example, take the quadruples $(7,8,9,10)$ and $(10,11,12,13)$. In the first case, $7 \times 10 = 70$ and $8 \times 9 = 72$. In the second case, $10 \times 13 = 130$ and $11 \times 12 = 132$. In both cases we see a difference of 2, as expected.

- Look for some more multiplicative properties that quadruples of consecutive integers have.
- Take any five consecutive numbers a, b, c, d, e in the sequence. What interesting multiplicative properties can you spot that are *always* true?
- Take any six consecutive numbers a, b, c, d, e, f in the sequence. What interesting multiplicative properties can you spot that are *always* true?

We will return to some of these questions in the next issue (January 2025).

Bamboo Baskets and Math

Article contributed by Padmapriya Shirali

I have always been very fond of baskets — palm leaf baskets, bamboo baskets, grass baskets. I love their texture, their earthy colour and their intricate weaves. They fascinate me!



(a)



(b)

Figure 1

My first encounter with baskets as a child was with a palm leaf basket that came along with models of a wooden kitchen set. Later, I encountered

them as 'Dhobi baskets' for storing used clothes, at Rishi Valley.

At Madanapalle, I used to visit the bamboo workers market which sold vegetable baskets, frames for sericulture, waste baskets, and various other objects made with bamboo. Trips to Varanasi exposed me to more and more varieties of bamboo objects and different types of weaves.



(a)



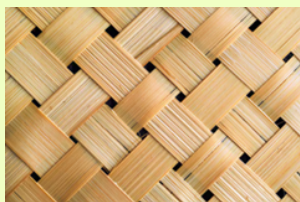
(b)

Figure 2

Most of us have access to these objects at home or in schools. If we don't, it is high time that we begin to replace plastic objects with these earth friendly materials!

Being a mathematics teacher, I naturally look for patterns, shapes and embedded geometric and mathematical concepts in basket making. Here are some ways of exploring these objects.

The first step is to identify the repeating unit.



(a)



(b)

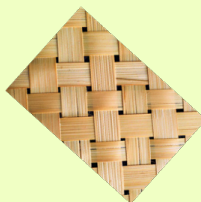
Figure 3: Source: <https://www.freepik.com/free-photos-vectors/bamboo-weave>

The repeating unit in this picture is a square shape. How will the students describe this unit? It is square in shape, with four overlapping strips. Do they notice how the overlap has been created? What steps of weaving will produce this unit?

Students can use coloured paper strips to produce the repeating unit. They can also describe the arrangement using numbers. If the pattern is turned, so that the strips become horizontal the pattern can be easily noted with numbers as “1 over and 1 under.” In standard form it is a ‘1 – 1’ pattern.



(a)



(b)

Figure 4

It is helpful for the students to model this pattern on grid paper using two different colours to represent the strips.



Figure 5: Grid drawing of the strips

Many questions can be raised and explored. For example: How many strips are needed to create the given design? At what angles do the overlapping strips meet? What type of symmetry do we find in the unit?



Figure 6: Three strips overlapping each other

Students can next explore how three strips can overlap each other. What is the shape that forms between the three strips? Is it a regular shape?

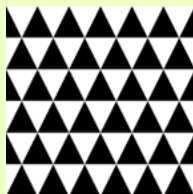


Figure 7

How many different triangles are possible? At what angles will the three overlapping strips meet? What type of symmetry will such a unit have or not have? How many of them tessellate? Why? Why not? They can follow it up with a grid drawing.

Is it possible to form a rhombus shape between 4 overlapping strips? Will such a unit tessellate? Why? Why not?



Figure 8

Students can create grid drawings and write the number pattern for each of these shapes.



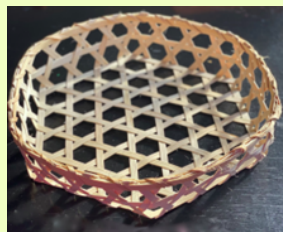
Figure 9

Is it possible to create any other quadrilateral with four overlapping strips? Will these shapes tessellate? Why? Why not?

Hexagonal shapes are frequently seen in basket weaving. What is the number pattern of a regular hexagon? Will it have diagonal patterns? Can the students generate the number pattern without a grid drawing?



(a)



(b)

Figure 10



(a)



(b)



(c)

Figure 11: Source for (c): <http://www.woodbambooplate.com/Twill%20WeaveCane%20Webbing.html>

Often in the past, wooden chairs and sofa seats used to be set with octagonal shaped weaves. Perhaps this was an appropriate design for a hot country like India.

Students can create the unit and study its symmetry. They can make grid drawings to see how the pattern tessellates. At what angles do the strips meet? Extend the pattern to see how it looks.

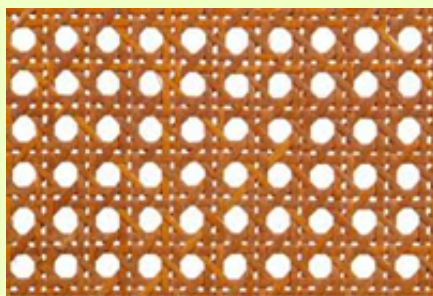
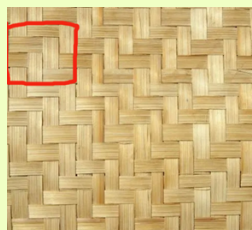


Figure 12

Let us now look at another bamboo weave pattern.



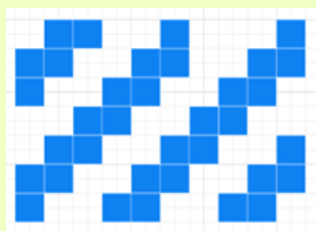
(a)



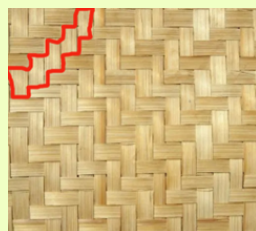
(b)

Figure 13

What is the unit here? There is a staggering of one vertical strip (Y axis / warp) in each row. This is a twill '2 – 2' pattern. Here is the grid drawing for this pattern.



(a)



(b)

Figure 14

Various diagonal patterns can be seen in this weave. Students can make the grid pattern for these staggered weaves.

Questions: Is the pattern made by the weave straight? Does it create a parallel pattern? How do you know? Are there any pattern lines that are perpendicular to the horizontal pattern? How do you know? Do the diagonal lines slope in the same direction?



(a)



(b)

Figure 15

Students can explore more such patterns and create some interesting weaves.



(a)

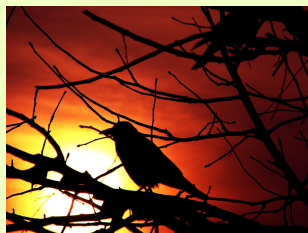


(b)

Figure 16

About the author ...

Padmapriya Shirali has been a math educator for the past four decades, teaching at various levels from Class 1 to Class 10. She has a strong interest in math pedagogy and conducting explorations in maths. She is a member of the NCERT textbook committee. She has worked closely with Shri P.K. Srinivasan, an eminent math educationist, and was part of the Rishi Valley Rural Education team which developed the multi-grade learning material known as 'school in a box.' She may be contacted at padmapriya.shirali@gmail.com

CROWS CAN COUNT!

"Crows at sunset in Saiee Park." Photograph by Behrooz Rezvani. Source: https://commons.m.wikimedia.org/wiki/File:Tree_branches,_crows_at_sunset_in_Saiee_Park_-_panoramio.jpg

Did you know that ... "crows have counting capacities near those of human toddlers who are beginning to develop a knack for numbers" and that "Crows do not appear to be capable of symbolic counting ... Instead, the birds are able to count by controlling the number of vocalizations they produce to correspond to associated cues ..." For details. see <https://www.scientificamerican.com/article/crows-rival-human-toddlers-in-counting-skills/>

Words of Wisdom from Prof. X

Article contributed by A. Ramachandran

Here are some “words of wisdom” from Prof. X! Do you agree with all the Professor’s claims?

1. Above average

Prof. X. claims that all his students scored above-average marks in a test.

2. World’s population in the distant past

Prof. X. muses, “I have two parents, four grandparents, eight great-grandparents, sixteen great-great-grandparents, and so on. So, the world’s population must have been higher in the olden days.”

3. Proof of the Pythagorean theorem

Prof. X. offers the following proof of the Pythagorean theorem. In $\triangle ABC$ (Figure 17), angle C is a right angle. We know that $\sin^2 A + \cos^2 A = 1$.

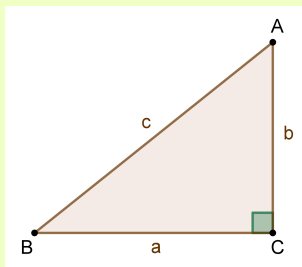


Figure 17

Now,

$$\sin A = \frac{BC}{AB} \quad \text{and} \quad \cos A = \frac{AC}{AB},$$

$$\therefore \frac{BC^2}{AB^2} + \frac{AC^2}{AB^2} = 1,$$

therefore $AB^2 = BC^2 + AC^2$. Hence proved!

About the author ...

A Ramachandran has had a long-standing interest in the teaching of mathematics and science. He studied physical science and mathematics at the undergraduate level and shifted to life science at the postgraduate level. He taught science, mathematics and geography to middle school students at Rishi Valley School for two decades. His other interests include the English language and Indian music. He may be contacted at archandran.53@gmail.com

Playing with Primitive Pythagorean Triples – 2

We continue our exploration of Primitive Pythagorean Triples or PPTs. In the July 2024 issue we learnt, among other things, that:

- The only PPT consisting of three consecutive numbers is $(3,4,5)$. Indeed, this is the only PPT consisting of three numbers in arithmetic progression.
- If we consider triangles whose sides are three consecutive numbers, we find that $(2,3,4)$ yields an obtuse-angled triangle; $(3,4,5)$ yields a right-angled triangle; and all other cases yield acute-angled triangles. (We left this statement for you to justify.)
- There are infinitely many PPTs (a,b,c) with $c = b + 1$, i.e., PPTs of the form $(a,b,b + 1)$; for example, $(5,12,13)$. Moreover, there is an easy way of generating all of them (which we described).
- There are infinitely many PPTs (a,b,c) with $c = b + 2$, i.e., PPTs of the form $(a,b,b + 2)$; for example, $(8,15,17)$. Though it is not part of

the description, it happens that b is odd for all such PPTs. Moreover, there is an easy way of generating all of them (which we described).

Continuing from the last two items in the list, let us examine the family of PPTs of the form $(a, b, b + 4)$.

PPTs where the last two numbers differ by 4

We now study Pythagorean triples of the form $(a, b, b + 4)$. The first step is to show that if $(a, b, b + 4)$ is a PPT, then b is odd.

More generally, we shall show that if (a, b, c) is a PPT, then b and c cannot *both* be even. We argue by contradiction. For, if both b and c are even, then from the relation $a^2 + b^2 = c^2$ it follows that a is even as well. (Do you see why?) But this means that all three numbers in the triple are even and therefore share the common prime factor 2. This contradicts the definition of a primitive triple! Hence it cannot be that both b and c are even. This means that at least one of b, c is odd. (Note that this remark holds for all PPTs. If you think about it carefully, you will realize that we can say something more: c must be odd. More later about this.)

Now let us apply what we have just learnt to the triple $(a, b, b + 4)$. Since b and $b + 4$ have the same parity (they are both odd or both even), it must be that b is odd.

Next, as the triple is Pythagorean we obtain:

$$\begin{aligned}a^2 + b^2 &= (b + 4)^2, \\ \therefore a^2 &= 8b + 16.\end{aligned}\tag{1}$$

The last relation tells us that a^2 is a multiple of 8, and therefore that a is a multiple of 4. (It cannot be that a is even but not a multiple of 4. Please convince yourself of this.) Let $a = 4m$, where m is some positive integer. Then we get:

$$16m^2 = 8b + 16, \quad \therefore b = 2m^2 - 2.$$

We had noted above that b must be odd; but the above relation tells us that b is even (as it is twice some number). It seems that we are contradicting ourselves!

This self-contradiction reveals something most unexpected! We shall state it as a theorem as it comes as such a surprise.

Theorem. *There are no PPTs of the form $(a, b, b + 4)$.*

Having bumped into this fact in an unplanned way, let us continue the exploration.

We remark before continuing that it is purely the algebra that has revealed this fact to us. "Algebra is kind; she gives us more than what we ask."

PPTs where the last two numbers differ by 3

What about triples where the last two numbers differ by 3? Let's explore this possibility carefully. This time we should be prepared for any finding!

Before proceeding we make a general statement applicable to any PPT (a, b, c) . Namely: *At most one out of a, b, c is a multiple of 3.* To see why, we argue by supposing that this is not the case. Suppose that both a and b are multiples of 3. Then since $c^2 = a^2 + b^2$, it follows that c too is a multiple of 3. But then all three of a, b, c are multiples of 3, contradicting the definition

of a PPT. So, it cannot happen that both a and b are multiples of 3. Exactly the same logic covers the remaining possibilities. (*Remark.* There is nothing unique about the number 3 in this paragraph. If we replace 3 by some other prime number p , the statement remains true.)

Let the PPT be $(a, b, b + 3)$. Then we have:

$$\begin{aligned}a^2 + b^2 &= (b + 3)^2, \\ \therefore a^2 &= 6b + 9.\end{aligned}\tag{2}$$

Since $6b + 9 = 3(2b + 3)$ is a multiple of 3, it follows that a^2 is a multiple of 3 and, therefore, a too is a multiple of 3. Let $a = 3u$ for some integer u . This leads to:

$$\begin{aligned}6b + 9 &= 9u^2, \\ \therefore 2b &= 3u^2 - 3.\end{aligned}$$

Since $3u^2 - 3$ is a multiple of 3, it follows that $2b$ is a multiple of 3 and, therefore, b too is a multiple of 3. Therefore, both a and b are multiples of 3, and this means that c too is a multiple of 3. But then, $(a, b, b + 3)$ is not a PPT at all!

Once again we have come against a self-contradiction. The contradiction shows that PPTs of the form $(a, b, b + 3)$ do not exist.

PPTs where the last two numbers differ by 5

What about triples where the last two numbers differ by 5? After the last two experiences we may be nervous about the outcome of our exploration!

We note at the start that for any PPT (a, b, c) , *at most one out of a, b, c is a multiple of 5*. This can be justified just like we did earlier, for the divisor 3.

Let the PPT be $(a, b, b + 5)$. Then we have:

$$\begin{aligned} a^2 + b^2 &= (b + 5)^2, \\ \therefore a^2 &= 10b + 25. \end{aligned} \tag{3}$$

Since $10b + 25$ is a multiple of 5, it follows that a^2 is a multiple of 5 and, therefore, a too is a multiple of 5. Let $a = 5u$ for some integer u . This leads to:

$$\begin{aligned} 10b + 25 &= 25u^2, \\ \therefore 2b &= 5u^2 - 5. \end{aligned}$$

Since $5u^2 - 5$ is a multiple of 5, it follows that $2b$ is a multiple of 5 and, therefore, b too is a multiple of 5. Therefore, both a and b are multiples of 5, and this means that c too is a multiple of 5. But then, $(a, b, b + 5)$ is not a PPT at all!

Yet again we have collided with a self-contradiction. It means that PPTs of the form $(a, b, b + 5)$ do not exist.

What next?

Well! We have uncovered some very surprising facts and we need to take stock of what we have found:

- There are infinitely many PPTs of the form $(a, b, b + 1)$ and $(a, b, b + 2)$.
(In the July 2024 issue we described ways of generating *all* of them.)

- There are **no** PPTs of the forms $(a, b, b + 3)$, $(a, b, b + 4)$, $(a, b, b + 5)$.

We now begin to wonder: are there any PPTs of the form $(a, b, b + 6)$? What about the form $(a, b, b + 7)$? What is so special about the numbers 3, 4, 5?

Continuing, there are obviously *lots* of questions we can ask. We will take up some of these in the next issue, but these are questions that you can explore on your own during the coming weeks.

And you may be able to pose many more such questions for yourself. Happy questioning!

Problems for you to try

1. Show that in any PPT (a, b, c) ,
 - (a) One out of a, b is odd while the other is even;
 - (b) c is odd.
2. Show that there are no PPTs with 6 as one of the numbers.
3. Show that there are no PPTs with 10 as one of the numbers.
4. Find other numbers of the kind that featured in the previous two problems — numbers which do not belong to any PPT.

Article contributed by Shailesh Shirali

Puzzles to Puzzle You – 2

The answers to these cryptarithms are given on page 164. Solutions will be given in the next issue of the magazine.

Puzzles for the October 2024 issue

1. Using the digits 2,0,2,4 once each and no other digits, and any of the standard mathematical operations ($+$, $-$, $\times$, $\div$, $\sqrt{}$, to the power of, brackets, factorial, ...), create expressions that yield in turn all of the numbers from 1 to 100. For example, the following expressions yield 1, 2 and 3 respectively:

$$(2 + 0 + 2) \div 4 = 1,$$

$$(2 + 0 + 2) \div \sqrt{4} = 2,$$

$$-(2 + 0)/2 + 4 = 3.$$

If possible, use the digits 2,0,2,4 in that order.

Find out how far you can continue this chain!

2. I keep as pets a few dogs, a few cats, a few rabbits, and a few ducks. Nine of my pets aren't dogs, ten aren't cats, eleven aren't rabbits, and twelve aren't ducks. How many pets do I have of each kind?
3. Three friends go to a restaurant and order a meal that costs ₹300. They each contribute ₹100. Later, the waiter realizes there's a mistake — the meal only costs ₹250. The waiter gives back ₹50. The three friends each take ₹10, and the waiter keeps ₹20 as a tip. Now, each friend has spent ₹90, which totals ₹270. The waiter kept ₹20, which totals ₹290. Where is the missing ₹10?
4. Solve the following cryptarithm:

$$TWO \times TWO = THREE.$$

Solutions to the cryptarithms in the July 2024 issue

Solution to Crazy Cryptarithm – 1

Solve the following cryptarithm:

$$\begin{array}{r} A \ B \ C \ D \\ \times \qquad \qquad 4 \\ \hline D \ C \ B \ A \end{array}$$

We argue in the following manner.

- A is even, as it is the units digit of a multiple of 4. Also, $ABCD < 2500$ (since $ABCD \times 4$ is a 4-digit number), so $A \leq 2$. Hence $A = 2$.
- From $ABCD \geq 2000$, we get $DCBA \geq 8000$, hence $D \geq 8$. Since $D \times 4$ has units digit 2, it follows that $D = 3$ or 8. Hence $D = 8$.

- Since $ABCD \times 4 < 9000$, it follows that $B \leq 2$.
- The multiplication $D \times 4$ yields a 'carry' of 3 (an odd number). Also, $C \times 4$ is even. Together, these imply that B is odd. Hence $B = 1$.
- Since $4C + 3$ has units digit 1, we get $C = 2$ or 7. The first possibility does not work out since $21C8 \times 4 \geq 8400$. Hence $C = 7$.
- Therefore $ABCD = 2178$. The multiplication now works out:

$$\begin{array}{r}
 2178 \\
 \times \quad 4 \\
 \hline
 8712
 \end{array}$$

Solution to Crazy Cryptarithm – 2

Solve the following cryptarithm:

$$\begin{array}{r}
 AB \\
 + \quad BC \\
 + \quad CA \\
 \hline
 ABC
 \end{array}$$

- Since $AB + BC + CA = ABC$, it follows that $AB + CA = A00$.
- The sum of two 2-digit numbers cannot exceed 199, so $A = 1$.
- Since $B + A$ is a multiple of 10, it follows that $B = 9$.
- Hence the answer is $19 + 98 + 81 = 198$.

Solution to Crazy Cryptarithm – 3

Solve the following cryptarithm, given that $A = 1$:

$$\begin{array}{r} B \ A \ D \\ \times \quad \quad 9 \\ \hline G \ O \ O \ D \end{array}$$

- Since $D \times 9$ has units digit D , it follows that $D = 0$ or 5 .
- Suppose $D = 0$. Since $A = 1$ (given), we get $O = 9$. Hence $B1 \times 9 = G99$. For $G99$ to be a multiple of 9 , we must have $G = 9$. But then $B1 \geq 111$. This is clearly not possible. Hence $D \neq 0$. Therefore $D = 5$.
- The cryptarithm now has the appearance: $B15 \times 9 = GOO5$. From this we get $O = 3$ (since $5 \times 9 = 45$, 'carry' 4 , and now: $9 \times 1 + 4 = 13$, 'carry' 1 .) Now we have: $B15 \times 9 = G335$. Hence $B \times 9$ has units digit 2 , giving $B = 8$.
- The final result: $815 \times 9 = 7335$.

Solution to Crazy Cryptarithm – 4

Solve the following cryptarithm:

$$\begin{array}{r} R \ O \ U \ N \ D \\ \times \quad \quad \quad 3 \\ \hline S \ Q \ U \ A \ R \ E \end{array}$$

We regret to say that *we do not have a satisfactory solution to this problem.*

If you have a satisfactory solution, please share it with us!

So, we shall simply state the final answer which we have obtained by “brute force” (with the help of a computer). We shall come back to this problem in a later issue when we have a completely satisfactory solution.

$$\begin{array}{r}
 3\ 5\ 7\ 4\ 6 \\
 \times \qquad\qquad 3 \\
 \hline
 1\ 0\ 7\ 2\ 3\ 8
 \end{array}$$

It is remarkable that this cryptarithm has a unique solution.

The Big Wide World of Data

Visualisation – 2

As stated in the earlier issue, this column aims to present a running series of short articles on the theme of data visualisation. We shall present a few striking examples of this in each issue (with little or no commentary).

There's a Lot More Junk Floating Around Earth Than There Used to Be

From: <https://sherwood.news/world/the-amount-of-space-junk-is-a-growing-issue/>. Authors: Tom Jones, David Crowther

Space oddities

Another day, another “cloudy with a chance of space debris” forecast in North America: a farmer in rural Canada recently found an 88-pound, 2-meter-wide hunk of charred metal that astronomy experts believe is from

a SpaceX rocket that shed the fragment in February. He plans to sell it to raise money for a hockey rink.

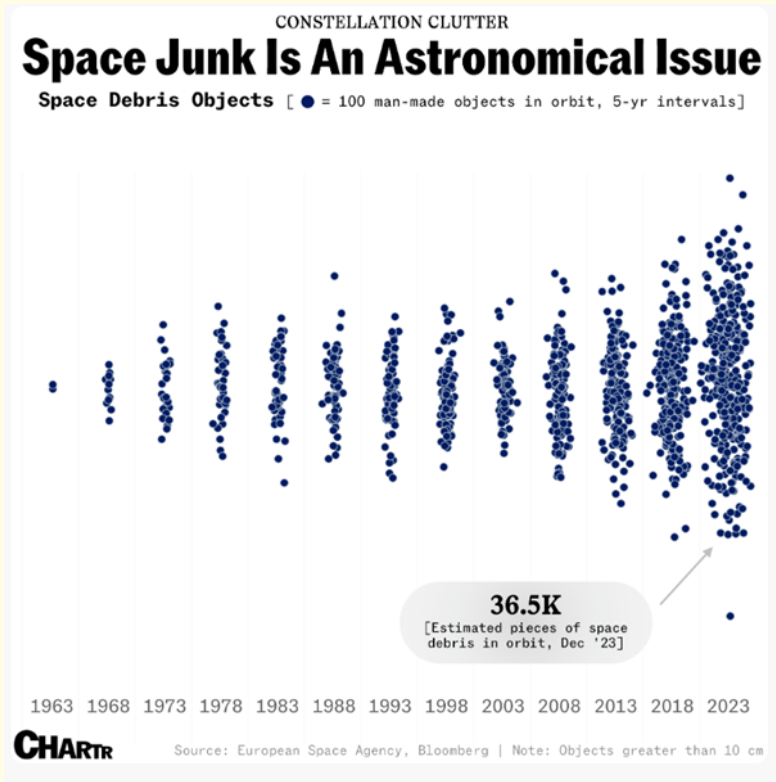


Figure 18: Space junk

The Canadian discovery follows news of a much smaller piece of space junk crashing through the roof of a Florida home in early March. They’re part of a growing phenomenon, with 200-400 man-made objects re-entering

Earth's atmosphere on average each year, most of which are mission-related debris from spacecraft, particles from disintegrations, and paint flecks... or dead satellites that continue to orbit the Earth long after we can still make use of them.

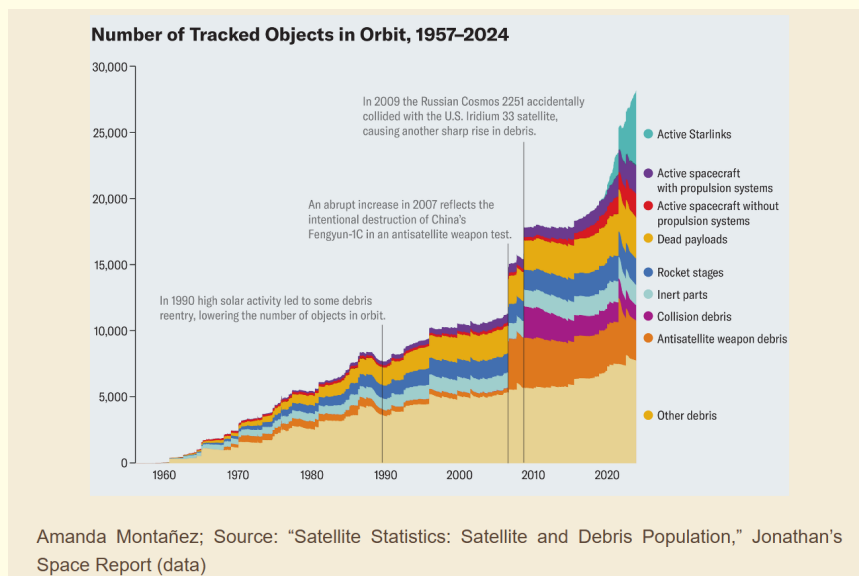


Figure 19: Space is becoming as crowded as our cities! Source: <https://www.scientificamerican.com/article/there-is-too-much-trash-in-space/>

According to the BBC, there's only one known case of someone being hit by falling space debris — an Oklahoma resident was harmlessly hit on the shoulder in 1997 — but the risk is real even if highly unlikely: the European

Space Agency (ESA) reports that space agencies and nation states accept a 1-in-10,000 chance of a casualty from a single uncontrolled re-entry.

Those stats may pick up in the coming years though, as the ESA tracks the ever-growing number of man-made objects that clutter the space around Earth. At the end of last year, a staggering 36,500 space debris objects over 10 cm in length were orbiting the Earth — perhaps little shock to anyone familiar with the Kessler Syndrome, a concerning theory that the more space junk there is, the more collisions there will be, causing a self-perpetuating chain reaction that could result in Earth's orbit becoming essentially unusable.

News for chess buffs

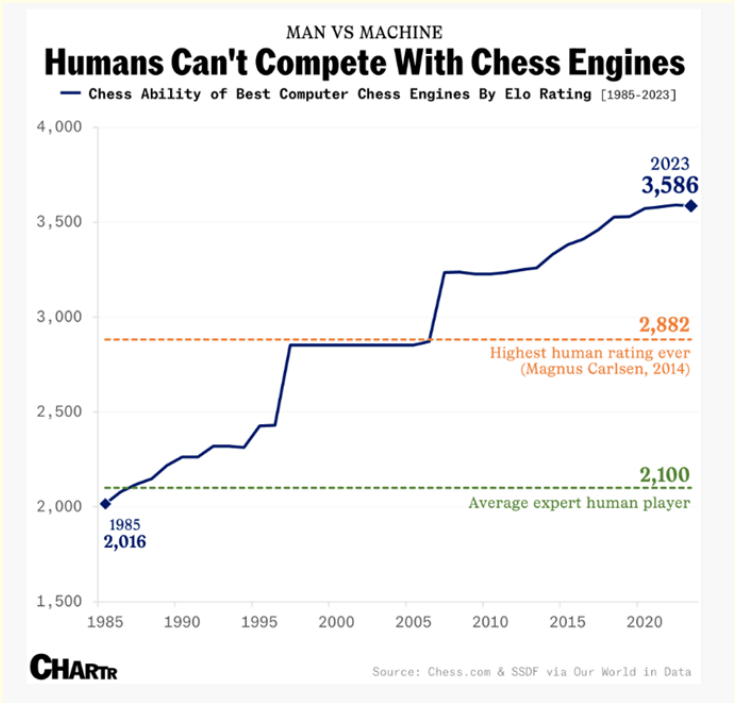


Figure 20: Chess engines have become incredibly powerful in recent times.
Source: https://www.linkedin.com/posts/andrew-rosca_activity-7205682935180480513-F5Hd/

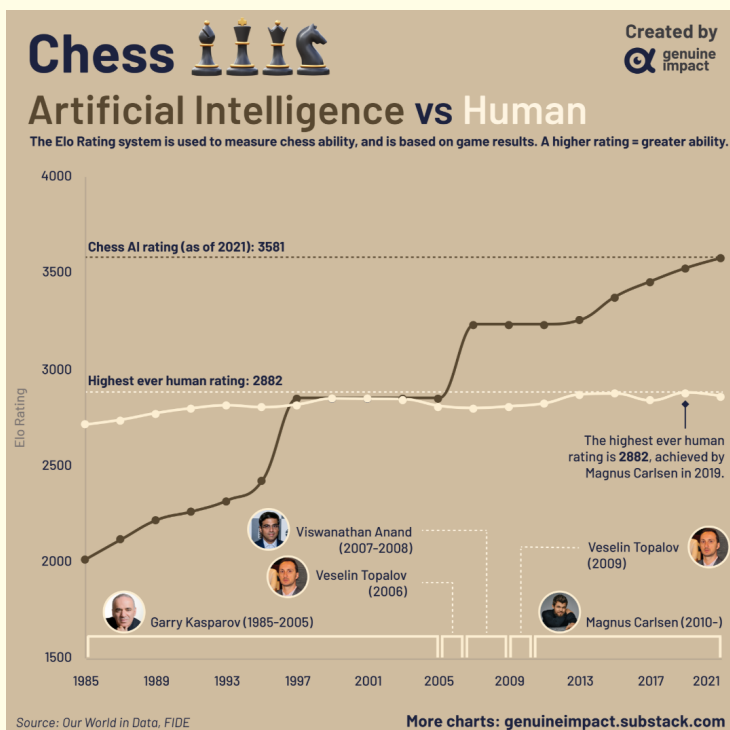


Figure 21: Chess engines have become ever more powerful in recent times.

Source: https://www.reddit.com/r/dataisbeautiful/comments/113ml18/oc_ai_vs_human_chess_elo_ratings_over_time/

How much energy does AI use?

From an article by Carole-Jean Wu in *Nature*, 21 August 2024. Source:

<https://tinyurl.com/2x3qer19>

Light bulbs have energy ratings – so why can't AI chatbots?

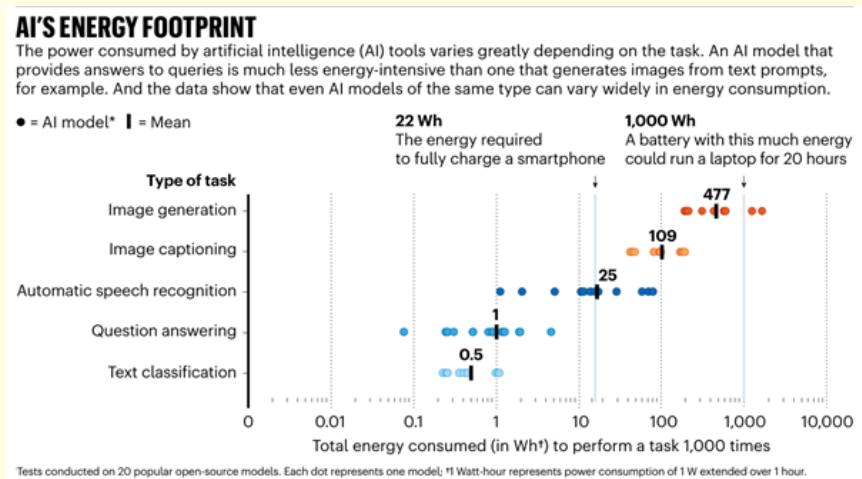


Figure 22: A scatter chart showing the energy consumed by various models for tasks such as image generation and speech recognition. The x -axis unit is watt-hour. Image generation consumes the most energy; the average is similar to a laptop running for 20 hours. Source: AI Energy Star project.

<https://www.nature.com/articles/d41586-024-02680-3>

There is a growing fear AI systems are using too much energy. Researchers have written about this in the magazine *Nature*. They wonder why we

cannot have a system whereby energy usage figures are made public just as they are for products like light bulbs. They suggest that this “can help users to choose the most appropriate models” and at the same time “encourage model developers to consider energy use as an important parameter.”

From the quoted *Nature* article: “As millions of people increasingly use generative artificial intelligence (AI) models for tasks ranging from searching the Web to creating music videos, there is a growing urgency about minimizing the technology’s energy footprint.

The worrying environmental cost of AI is obvious even at this nascent stage of its evolution. A report published in January by the International Energy Agency estimated that the electricity consumption of data centres could double by 2026, and suggested that improvements in efficiency will be crucial to moderate this expected surge.

Some tech-industry leaders have sought to downplay the impact on the energy grid. They suggest that AI could enable scientific advances that might result in a reduction in planetary carbon emissions. Others have thrown their weight behind yet-to-be-realized energy sources such as nuclear fusion.

However, as things stand, the energy demands of AI are keeping ageing coal power plants in service and significantly increasing the emissions of companies that provide the computing power for this technology. Given that the clear consensus among climate scientists is that the world faces a ‘now or never’ moment to avoid irreversible planetary change, regulators, policymakers and AI firms must address the problem immediately. ...”

How To Prove It – 2

We continue our series of articles on the theme of proof. In the July 2024 issue we featured a well-known proof technique – *proof by induction*. We shall continue our discussion on this technique.

Experimentation and the process of generalization

Experimentation in the sciences is well-known, but in the popular mindset, there is no parallel of this in mathematics. Perhaps this impression has come about because the subject is dominated by logical reasoning and formal proofs. But it is a misunderstanding of a very basic approach to questioning and inquiry in mathematics.

In reality, experimentation plays a vital role in mathematical research. Experimentation and data analysis form the foundation for conjecturing. In the sciences, that same step is known as *hypothesizing*. But it amounts to the same thing.

Mathematicians frequently use experiments to build a database about a suspected pattern, or to test out an existing conjecture. They also use

experiments to build an intuitive understanding of a subject; by examining specific instances, they begin to form a picture of the general pattern. A crucial component of this is the *analysis of data*. For example if the subject of study is the distribution of prime numbers, a mathematician may look at an existing database of prime numbers (or use computer software to build such a database) and then apply statistical and other computational techniques to the data. Looking for trends and relationships in data is an art form in itself; it forms a central part of data analysis. (We will explore this theme later.) Such analyses can reveal unexpected patterns, leading to the formulation of a *conjecture*. Many of the currently unsolved problems in mathematics (also called “open problems”) have come about as the result of such experimentation. Later in this article we shall give examples of conjectures that emerge in this manner.

A conjecture could also be called a piece of guesswork! But it is not just random guesswork — it has plenty of evidence to back it up.

A famous example of this is the *prime number theorem* which concerns the distribution of prime numbers. Towards the end of the 18th century, mathematicians noticed some striking patterns in the published data (by Legendre and Gauss, independently); there seemed to be a relationship between prime numbers and (of all things) logarithms! Several decades passed before this conjectured relationship was proved.

A more playful example of this is the (still open) Collatz Conjecture, also called the “ $3n + 1$ problem”. It arose when mathematicians observed a certain numerical phenomenon happening repeatedly.

While experimentation, data analysis and conjecturing are a crucial part of mathematics, we may describe them as one half of the subject. The other half is the development of logical arguments to establish the truth of the

conjecture (or, as has happened in a few cases, to disprove it). Without this essential second half, the subject remains incomplete.

It is important that we keep this perspective in mind as it has become an increasingly dominant theme in current times with the easy availability of powerful computers.

Examples of conjectured relationships

Consider the following situations.

1. Say we are studying the powers of 2. On observing the following statements,

$$1 + 2 = 2^2 - 1, \quad 1 + 2 + 2^2 = 2^3 - 1, \quad 1 + 2 + 2^2 + 2^3 = 2^4 - 1,$$

we may be led to guess that $1 + 2 + 2^2 + \cdots + 2^{n-1} = 2^n - 1$ for all positive integers n .

2. Write $f(n)$ for the sum $1^2 + 2^2 + 3^2 + \cdots + n^2$. Say we are trying to discover a formula for $f(n)$. Let $g(n) = 1 + 2 + 3 + \cdots + n$. We generate a list of values $f(n)$ and $g(n)$ for several values of n and display the data as follows:

n	1	2	3	4	5	6	7	8	9	10
$f(n)$	1	5	14	30	55	91	140	204	285	385
$g(n)$	1	3	6	10	15	21	28	36	45	55

We notice that $f(n)$ is a multiple of $g(n)$ for certain values of n . On noticing this, we generate a table of values of $f(n)/g(n)$, to see if this

will reveal some nice pattern. It does! —

n	1	2	3	4	5	6	7	8	9	10
$\frac{f(n)}{g(n)}$	1	$\frac{5}{3}$	$\frac{7}{3}$	3	$\frac{11}{3}$	$\frac{13}{3}$	5	$\frac{17}{3}$	$\frac{19}{3}$	7

This becomes *much* more revealing if we write it as follows:

n	1	2	3	4	5	6	7	8	9	10
$\frac{3f(n)}{g(n)}$	3	5	7	9	11	13	15	17	19	21

The pattern is now crystal clear and it is an easy task to guess a formula for $f(n)/g(n)$, and therefore a formula for $f(n)$ as we already know a formula for $g(n)$.

3. Encouraged by our success, let us do something similar for the sum $a(n) = 1^4 + 2^4 + 3^4 + \cdots + n^4$. We wish to find a formula for $a(n)$. Let $b(n) = 1^2 + 2^2 + 3^2 + \cdots + n^2$. We generate a list of values $a(n)$ and $b(n)$ for several values of n and display the data as follows:

n	1	2	3	4	5	6	7	8
$a(n)$	1	17	98	354	979	2275	4676	8772
$b(n)$	1	5	14	30	55	91	140	204

We notice (as earlier) that $a(n)$ is a multiple of $b(n)$ for some values of n (e.g., $98 = 14 \times 7$; $2275 = 91 \times 25$). On noticing this, we generate a table of values of $a(n)/b(n)$, to see if this will reveal some nice pattern:

n	1	2	3	4	5	6	7	8
$\frac{a(n)}{b(n)}$	1	$\frac{17}{5}$	7	$\frac{59}{5}$	$\frac{89}{5}$	25	$\frac{167}{5}$	43

As earlier, the pattern becomes clearer if we rewrite the table:

n	1	2	3	4	5	6	7	8
$\frac{5a(n)}{b(n)}$	5	17	35	59	89	125	167	215

Can we guess a formula for $a(n)$ now? Try!

- On examining numbers of the form $2^n + 1$, we notice quickly that the numbers $2^1 + 1 = 3$, $2^2 + 1 = 5$, $2^4 + 1 = 17$ and $2^8 + 1 = 257$ are all prime, but not $2^3 + 1 = 9$, $2^5 + 1 = 33$, $2^6 + 1 = 65$, ... We may now take a (huge!) leap and hypothesize that $2^n + 1$ is prime whenever n is a power of 2, and composite whenever n is not a power of 2. (Here, n is a positive integer.)
- While playing around with the sequence of primes (2, 3, 5, 7, 11, 13, ...), we may notice the following: if we form their partial sums starting from the first term, we get the numbers

$$2, 2 + 3 = 5, 5 + 5 = 10, 10 + 7 = 17, 17 + 11 = 28, 28 + 13 = 41,$$

and so on. If to these numbers we add, in alternation, the numbers 3 and 2 (that is, we add 3 to the first number, 2 to the second number, 3 to the third number, and so on), we obtain the sequence

$$5, 7, 13, 19, 31, 43, \dots$$

This sequence seems to contain only prime numbers! We may now want to take another big leap and hypothesize that this procedure generates only primes.

We can go on in this manner and list many more such examples; but the above should suffice.

A few illustrative problems

1. Prove that for all positive integers n , we have

$$1 \cdot n + 2 \cdot (n - 1) + 3 \cdot (n - 2) + \cdots + n \cdot 1 = \frac{n(n+1)(n+2)}{6}.$$

2. Prove that 6 is a divisor of $n(n+1)(2n+1)$ for all positive integers n .

3. Prove that $n^5 - n$ is a multiple of 5 for all positive integers n .

4. Prove that the Fibonacci numbers satisfy the inequality $f_n < (7/4)^n$ for all positive integers n .

5. Prove that $n! > 2^n$ for all positive integers $n > 3$.

We present the solutions below. Please study them carefully. Take careful note of the manner in which they have been written.

Solution of problem 1

Let functions f and g be defined as follows:

$$f(n) = 1 \cdot n + 2 \cdot (n - 1) + 3 \cdot (n - 2) + \cdots + n \cdot 1,$$

$$g(n) = \frac{n(n+1)(n+2)}{6}.$$

We first check the “early cases.” We have:

$$f(1) = 1, \quad g(1) = \frac{1 \cdot 2 \cdot 3}{6} = 1, \quad \therefore f(1) = g(1),$$

$$f(2) = 1 \cdot 2 + 2 \cdot 1 = 4, \quad g(2) = \frac{2 \cdot 3 \cdot 4}{6} = 4, \quad \therefore f(2) = g(2).$$

We must find a way of connecting $f(n+1)$ with $f(n)$, and $g(n+1)$ with $g(n)$.

We have:

$$f(n+1) = 1 \cdot (n+1) + 2 \cdot (n) + 3 \cdot (n-1) + \cdots + n \cdot 2 + (n+1) \cdot 1,$$

$$f(n) = 1 \cdot n + 2 \cdot (n-1) + 3 \cdot (n-2) + \cdots + n \cdot 1.$$

Term-by-term subtraction yields,

$$f(n+1) - f(n) = 1 + 2 + 3 + \cdots + n + (n+1) = \frac{(n+1)(n+2)}{2}.$$

Similarly,

$$g(n+1) = \frac{(n+1)(n+2)(n+3)}{6},$$

$$g(n) = \frac{n(n+1)(n+2)}{6},$$

and so

$$g(n+1) - g(n) = \frac{(n+1)(n+2)(n+3-n)}{6} = \frac{(n+1)(n+2)}{2}.$$

We see that $f(n+1) - f(n) = g(n+1) - g(n)$.

So, if $f(n) = g(n)$, then $f(n+1) = g(n+1)$ as well.

Since $f(1) = g(1)$, it follows that $f(n) = g(n)$ for all positive integers n . ■

Solution of problem 2

We must prove that 6 is a divisor of $n(n+1)(2n+1)$ for all positive integers n . The proposition is true for $n = 1$ since 6 is a divisor of $1 \cdot 2 \cdot 3$.

Let $f(n) = n(n+1)(2n+1)$.

We must find a way of connecting $f(n+1)$ and $f(n)$. We have:

$$\begin{aligned}f(n+1) - f(n) &= (n+1)(n+2)(2n+3) - n(n+1)(2n+1) \\&= (n+1) \cdot ((n+2)(2n+3) - n(2n+1)) \\&= (n+1) \cdot (2n^2 + 7n + 6 - 2n^2 - n) \\&= (n+1) \cdot (6n+6) = 6(n+1)^2.\end{aligned}$$

We see that the difference $f(n+1) - f(n)$ is a multiple of 6.

So, if $f(n)$ is a multiple of 6, then $f(n+1)$ too is a multiple of 6.

Since $f(1)$ is a multiple of 6, it follows that $f(n)$ is a multiple of 6 for all positive integers n . 

Solution of problem 3

We must prove that $n^5 - n$ is a multiple of 5 for all positive integers n . We follow the same strategy as in problem 2.

Let $f(n) = n^5 - n$.

Then $f(1) = 0$ is a multiple of 5.

We must now find a way of connecting $f(n+1)$ and $f(n)$. We have:

$$\begin{aligned} f(n+1) - f(n) &= \left((n+1)^5 - (n+1)\right) - \left(n^5 - n\right) \\ &= (n+1)^5 - n^5 - 1 \\ &= 5n^4 + 10n^3 + 10n^2 + 5n, \end{aligned}$$

by using the binomial theorem.

We see that the difference $f(n+1) - f(n)$ is a multiple of 5.

So, if $f(n)$ is a multiple of 5, then $f(n+1)$ too is a multiple of 5.

Since $f(1)$ is a multiple of 5, it follows that $f(n)$ is a multiple of 5 for all positive integers n . ■

Solution of problem 4

We must prove that the Fibonacci numbers satisfy the inequality $f_n < (7/4)^n$ for all positive integers n . For this we first recall the definition: $f_1 = 1$, $f_2 = 1$, and $f_n = f_{n-1} + f_{n-2}$ for $n > 2$.

Next, we check that the inequality holds at the start: $f_1 = 1 < 7/4$ (true); $f_2 = 1 < (7/4)^2$ (true).

Keeping the definition in mind, we shall assume that the stated inequality holds for *two* consecutive positive integers and then show that if this happens, it will be true for the next integer as well. That is,

$$\text{If } f_{n-2} < \left(\frac{7}{4}\right)^{n-2} \text{ and } f_{n-1} < \left(\frac{7}{4}\right)^{n-1} \text{ then } f_n < \left(\frac{7}{4}\right)^n.$$

But this is straightforward; we only have to use the definition.

We have:

$$\begin{aligned}
 f_n &= f_{n-1} + f_{n-2} \\
 &< \left(\frac{7}{4}\right)^{n-2} + \left(\frac{7}{4}\right)^{n-1} \\
 &= \left(\frac{7}{4}\right)^{n-2} \left(1 + \frac{7}{4}\right) \\
 &= \left(\frac{7}{4}\right)^{n-2} \cdot \frac{11}{4} \\
 &< \left(\frac{7}{4}\right)^{n-2} \cdot \left(\frac{7}{4}\right)^2, \quad \text{since } \frac{11}{4} < \frac{49}{16}.
 \end{aligned}$$

$$\text{Hence } f_n < \left(\frac{7}{4}\right)^n.$$

So if $f_{n-2} < (7/4)^{n-2}$ and $f_{n-1} < (7/4)^{n-1}$, then $f_n < (7/4)^n$.

As we have already checked that $f_1 < 7/4$ and $f_2 < (7/4)^2$, it follows that the inequality $f_n < (7/4)^n$ is true for all positive integers n . ■

Solution of problem 5

We must prove that $n! > 2^n$ for all positive integers $n > 3$.

The interesting feature here is that the inequality to be proved is *false* if we substitute $n = 1, 2$ or 3 ; please check! So the clause " $n > 3$ " is an essential part of the problem.

The statement is true for $n = 4$, since $4! = 24 > 2^4$.

Now we must show that if $n! > 2^n$ for some n , then $(n+1)! > 2^{n+1}$ as well. Using the assumed inequality, we obtain:

$$\begin{aligned}(n+1)! &= (n+1) \cdot n! \\ &> (n+1) \cdot 2^n && \text{since } n! > 2^n, \\ &> 2 \cdot 2^n && \text{since } n+1 > 2.\end{aligned}$$

Therefore $(n+1)! > 2^{n+1}$. We have shown that if $n! > 2^n$ then $(n+1)! > 2^{n+1}$ as well.

It follows that the stated inequality holds for all integers $n > 3$. ■

Closing remarks

We have covered different kinds of problems in these two articles on the use of induction in mathematical proofs: summation problems, divisibility problems, and inequalities. But this does not exhaust the full range of application of this approach. Proof by induction is far more versatile than one may suppose initially. We will come across its application quite frequently in these articles.

Proofs Without Words – 1

A *proof without words* (PWW for short) is a visual portrayal of a proof that would otherwise be expressed in ordinary language or algebra. It attempts to convey the central logic of a proof without using any symbols (or reducing their usage to a bare minimum). Such proofs make skilful use of diagrams and other visual aids. They can be aesthetically very pleasing. They need to be considered an art form!

A classic example of such a proof is Bhaskaracharya's "Behold!" proof for the Pythagorean theorem. The legend goes that he simply drew the diagram (Figure 23) and said, "Behold!"

The proof works by computing areas. Each outer triangle has area $\frac{1}{2}ab$, the inner square with side c has area c^2 , and the outer square with side $a + b$ has area $(a + b)^2$. Straightforward algebra now yields $c^2 = a^2 + b^2$.

Of course, all this is implicit; it is *assumed* that the reader will be able to look at the diagram and deduce all these area relationships mentally. Hence "Behold!" (Bhaskara clearly expects a lot from his readers!)

Another implicit assumption is that four congruent copies of a right-angled triangle can be fitted together in the manner shown, i.e., so that the inner

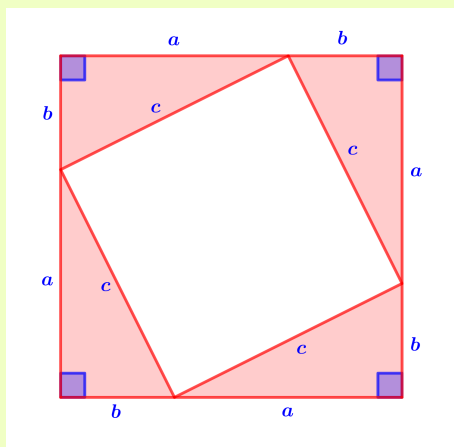


Figure 23: Bhaskara's "Behold!" proof for the Pythagorean theorem

figure is indeed a square, and the outer figure too is a square. Providing the necessary justification is easy, and the reader should do so. But the point is that it is implicit. The reader is *expected* to provide the details.

It might be wondered: to what extent is a PWW actually a proof? In the strict sense of the word a PWW must not be regarded as a proof; too much is left implicit, and too much is taken for granted. It may be better to regard a PWW as a *suggestion of a proof*, and PWWs as an art form.

What is undeniable is that PWWs have great educational value. So let us acknowledge the pleasure they bring us and be grateful that they exist!

We shall regularly showcase PWWs in this magazine. Two more PWWs are displayed on the next two pages, followed by some striking PWWs in support of the Pythagorean theorem.

Summation of an infinite GP with common ratio 1/4

Here is a beautiful composition by Rick Mabry (see [1]) which proves the equality (some might prefer to say that it *illustrates* the equality):

$$\frac{1}{4} + \left(\frac{1}{4}\right)^2 + \left(\frac{1}{4}\right)^3 + \left(\frac{1}{4}\right)^4 + \cdots = \frac{1}{3}.$$

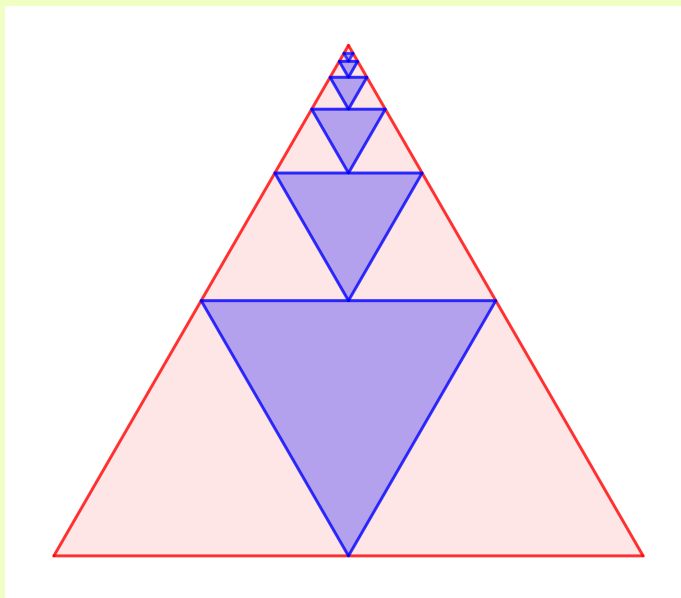
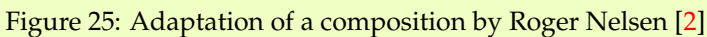


Figure 24: A composition by Rick Mabry of Louisiana State University

Readers should take note of the elegance and economy of this composition: absolutely no symbols or numbers appear anywhere in the diagram!

Given any two unequal, positive numbers a, b , their harmonic mean (HM), the geometric mean (GM) and arithmetic mean (AM) are defined as follows:

These three means satisfy the following order relationship: $HM < GM < AM$. The following composition adapted from Roger Nelsen (see [2]) proves these inequalities. (Here $a > b > 0$. If $a = b$, the means are all equal.)



Comments on Nelsen's PWW

The observant reader would have noticed that there is a fourth mean hiding in Figure 25. This is the *root-mean-square* (RMS):

$$RMS(a, b) = \sqrt{\frac{a^2 + b^2}{2}}.$$

Nelsen's composition proves (some would say "illustrates") the following string of inequalities: if $a, b > 0$ and $a \neq b$, then

$$HM(a, b) < GM(a, b) < AM(a, b) < RMS(a, b).$$

For example, take $a = 9$ and $b = 1$. Then we get:

$$HM = \frac{2 \cdot 9 \cdot 1}{9 + 1} = \frac{9}{5} = 1.8, \quad GM = \sqrt{9 \cdot 1} = 3,$$

$$AM = \frac{9 + 1}{2} = 5, \quad RMS = \sqrt{\frac{9^2 + 1^2}{2}} = \sqrt{41} \approx 6.403.$$

Further reading

- [1] Rick Mabry, "Proof without Words: $\frac{1}{4} + \left(\frac{1}{4}\right)^2 + \left(\frac{1}{4}\right)^3 + \cdots = \frac{1}{3}$," *Mathematics Magazine*, Vol. 72, No. 1 (Feb., 1999), p. 63; from <https://www.jstor.org/stable/2691318>
- [2] Roger Nelsen, "Proof without Words: The Harmonic Mean-Geometric Mean-Arithmetic Mean-Root Mean Square Inequality," *Mathematics Magazine*, Vol. 60, No. 3 (Jun., 1987), p. 158; from <https://www.jstor.org/stable/2689561>

Proofs-without-words for the Pythagorean theorem – 1

Article contributed by Adrian Kujur

Among the various approaches for proving the Pythagorean theorem (and there are numerous ways available), there are many that can be described as proofs without words or PWWs.

Some of these are variations of Bhaskara's "Behold!" proof; they are based on the algebraic expansion of $(a + b)^2$ or $(a - b)^2$. For example, the proof by James Garfield (who later became President of the USA) is of this type. **All such proofs are essentially algebraic in nature.**

Others are **proofs by dissection**: they describe how the two small squares on the legs of the right-angled triangle can be dissected and the pieces reassembled to make up the large square on the hypotenuse of the triangle.

In this article we present several such dissection schemes. *Each scheme works for a given ratio of the legs.*

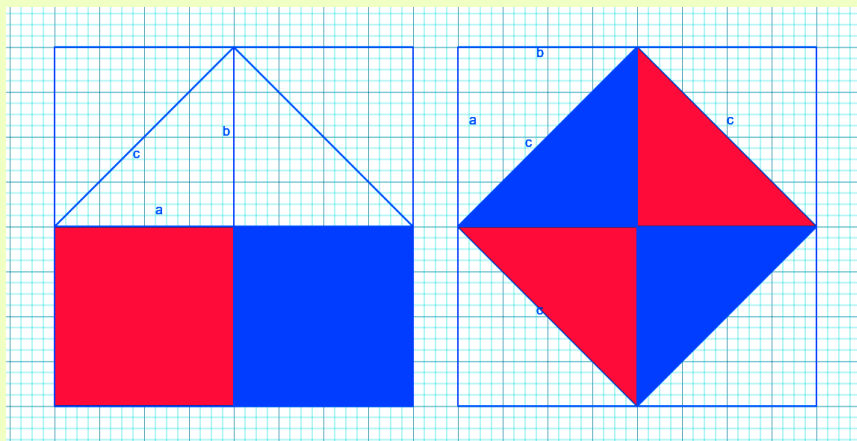
When the ratio of the legs is 1 : 1

Figure 26

This one is fairly straightforward.

It is easy to see how the individual squares on sides a and b have been cut into two pieces each and the pieces reassembled on the right to make up the square on side c .

When the ratio of the legs is 1 : 1 (a different scheme)

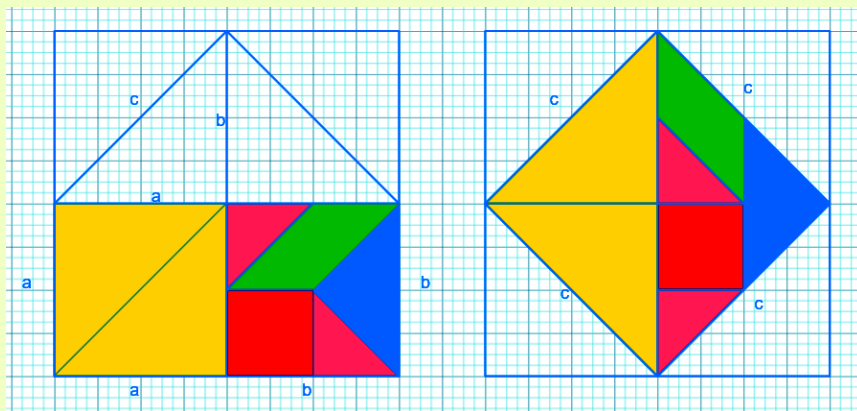


Figure 27

This one is more complex than the previous one.

Observe that the squares on sides a and b have been dissected in a different manner. The square on side a has been cut into just two pieces, whereas the square on side b has been cut into five pieces of different shapes.

The pieces have then been reassembled to make the square on side c .

As you can see, the reassembly is not at all obvious.

When the ratio of the legs is 1 : 2

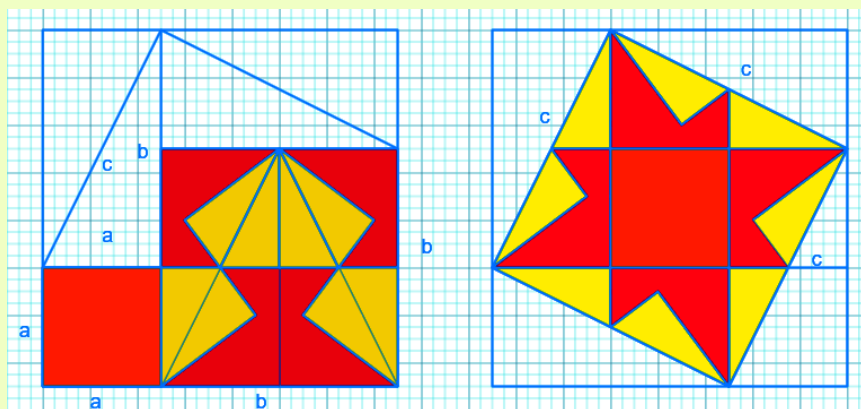


Figure 28

Here the ratio of the legs $a : b$ is 1 : 2.

The square on side a stays intact; it is not cut at all.

But the square on side b is divided into as many as 12 different pieces. Of these, 8 pieces are congruent copies of a smaller right-angled triangle whose ratio of legs is 1 : 2 (the same as for the original triangle). The remaining four pieces are congruent copies of a different shape.

The manner in which all these pieces fit together to make the square on side c is quite complex. The reader should study this scheme quite closely.

We will be continuing this article on PWWs for the Pythagorean theorem in subsequent issues of the magazine.

About the author ...

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He has been teaching Mathematics since 2010. At the same time he has been associated with the production of teaching-learning materials and running a teacher training programme in mathematics for many years. His teaching innovations have been recognised by Azim Premji Foundation in Ambikapur, Chhattisgarh.

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Gems From Geometry – 1

This is going to be a continuing series of articles on striking results (“gems”) from geometry. There is a vast collection of such gems, discovered over the past two millennia by generations upon generations of mathematicians and engineers — results about triangles, circles, quadrilaterals, and (in a few cases) polygons with more than four sides. We will describe them a few at a time over successive issues of the magazine.

We start with a (not very well-known) theorem about triangles; it has an interesting connection with a popular puzzle about buried treasure. In the next issue we shall study a theorem about quadrilaterals.

A pair of equal, perpendicular segments associated with a triangle

Associated with each side of a triangle, there are two equal, perpendicular segments. What makes this result appealing is the variety of approaches that can be used to prove it.

Theorem ("Buried treasure"). *Given any triangle ABC , let M denote the midpoint of side BC . Let squares $ABEH$ and $CAGH$ be drawn on the sides AB and AC of the triangle, lying outside the triangle. Let the centres of these two squares be P and Q respectively. Then PM and QM are equal and perpendicular to each other.*

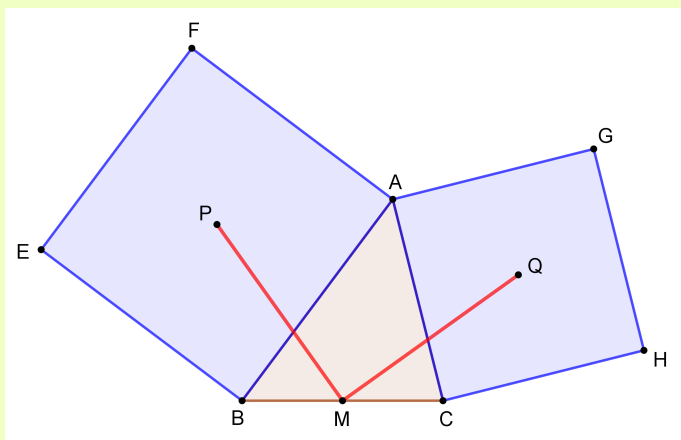


Figure 29: The "buried treasure" theorem: $PM = QM$ and $PM \perp QM$

We have nicknamed this the "Buried Treasure Theorem" as it is closely connected with a puzzle about buried treasure. As with many such results about triangles, we do not know who first discovered this result.

Proof of the buried treasure theorem

We offer two different proofs, one using familiar Euclidean geometry, and the other using complex numbers.

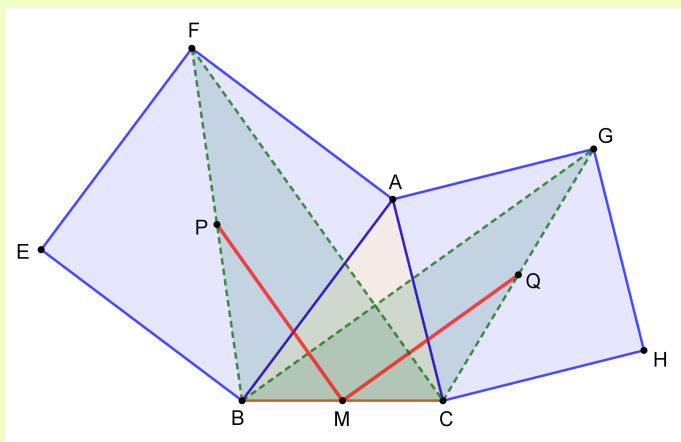


Figure 30: Geometric proof of the buried treasure theorem

Geometric proof. Draw segments BF , CF , BG , CG . Consider $\triangle BCF$ and $\triangle BCG$. We may now invoke the midpoint theorem, since M is the midpoint of BC , P is the midpoint of BF , and Q is the midpoint of CG . We conclude that (a) $PM \parallel FC$ and $PM = FC/2$, and (b) $QM \parallel GB$ and $QM = GB/2$.

Next, consider $\triangle AFC$ and $\triangle ABG$. We have $AF = AB$ and $AC = AG$ (sides of a square), and $\angle CAF = \angle GAB$ (both angles are equal to $90^\circ + \angle BAC$). Hence $\triangle AFC \cong \triangle ABG$ ("side-angle-side" congruence), therefore $FC = BG$.

Moreover, $AF \perp AB$ and $AC \perp AG$ (sides of a square). Hence $FC \perp BG$.

It follows that $PM = QM$ and $PM \perp QM$. ■

Consider the interplay of ideas in the proof above: we have made use of

the midpoint theorem, side-angle-side congruence, and rotation through a right angle. It makes for an interesting mosaic.

Proof using complex numbers We recall an extremely important property of the imaginary unit i : *In the complex plane, multiplication by i accomplishes a counter-clockwise rotation through a right angle.* (Here $i\sqrt{-1}$.) The figure shown here illustrates this property.

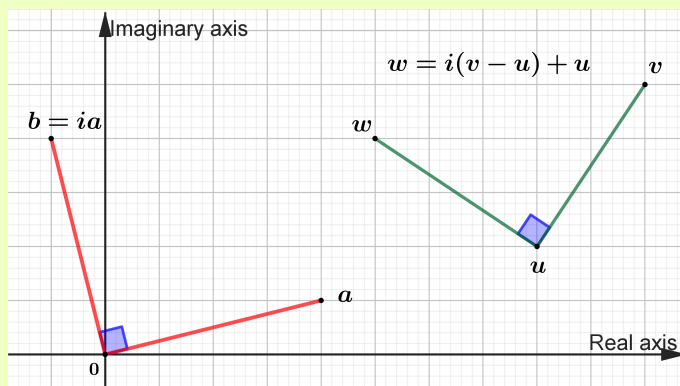


Figure 31

Embed the given configuration in the complex plane so that M lies at the origin and BC lies on the real axis. Let the complex number corresponding to each point be denoted by the corresponding lowercase letter; this label is also called the *affix* of that point. So the affix of A is a , the affix of B is b , etc. Note that $c = -b$.

Invoking the rotation property noted above, we obtain,

$$e = i(a - b) + b,$$

$$g = i(c - a) + a, \quad \therefore g = -i(b + a) + a.$$

We also have:

$$2p = a + e, \quad m = 0, \quad 2q = c + g.$$

The claims that $PM = QM$ and $PM \perp QM$ will get proved *simultaneously* (two proofs in one!) if we can show that $p = iq$; equivalently, if we can show that $2p = i \cdot 2q$. Now we have,

$$2p = a + b + i(a - b),$$

$$2q = (a - b) - i(a + b).$$

The relation $2p = i \cdot 2q$ follows from the above two equalities, since $i^2 = -1$.

Hence $PM = QM$ and $PM \perp QM$. ■

The pirate and the buried treasure

Now let us reveal the connection of the above theorem and the puzzle involving the pirate and buried treasure. Here is the exact statement of the puzzle from the classic book *One, Two, Three ... Infinity* by physicist George Gamow:

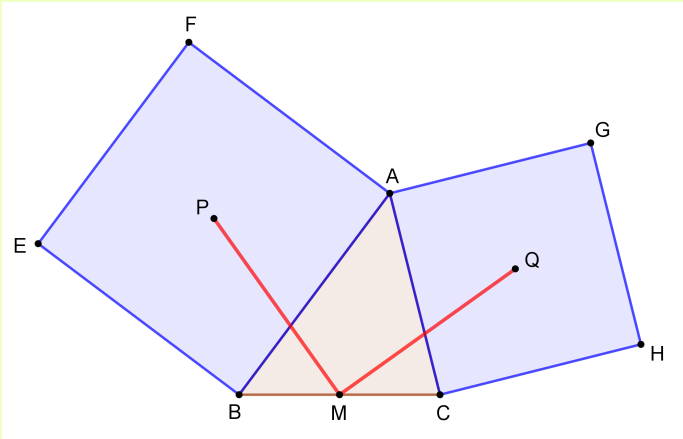


Figure 32: The “buried treasure” theorem: proof using complex numbers

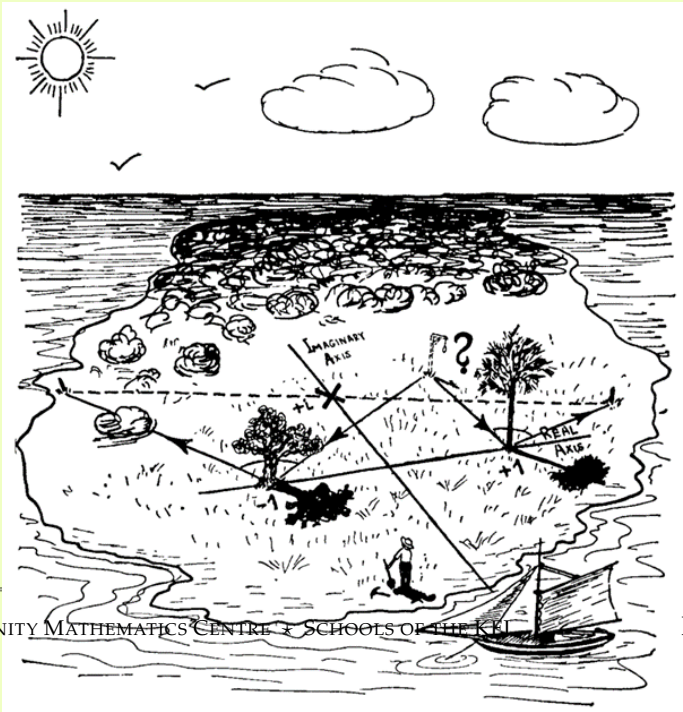


Figure 33: The pirate and the buried treasure. Source: George Gamow’s *One, Two, Three ... Infinity*

There was a young and adventurous man who found among his grandfather's papers a piece of parchment that revealed the location of a hidden treasure. The instructions read: "Sail to North latitude __ and West longitude __ where thou wilt find a deserted island. There lieth a large meadow, not pent, on the north shore of the island where standeth a lonely oak and a lonely pine. There thou wilt see also an old gallows on which we once were wont to hang traitors. Start thou from the gallows and walk to the oak counting thy steps. At the oak thou must turn *right* by a right angle and take the same number of steps. Put here a spike in the ground. Now must thou return to the gallows and walk to the pine counting thy steps. At the pine thou must turn *left* by a right angle and see that thou takest the same number of steps, and put another spike into the ground. Dig halfway between the spikes; the treasure is there."

The instructions were quite clear and explicit, so our young man chartered a ship and sailed to the South Seas. He found the island, the field, the oak and the pine, but to his great sorrow the gallows was gone. Too long a time had passed since the document had been written; rain and sun and wind had disintegrated the wood and returned it to the soil, leaving no trace even of the place where it once had stood. Our adventurous young man fell into despair, then in an angry frenzy began to dig at random all over the field. But all his efforts were in vain; the island was too big! So he sailed back with empty hands. And the treasure is probably still there.

(Gamow adds, to bring a feeling of authenticity to this story, "The actual figures of longitude and latitude were given in the document but are omitted in this text, in order not to give away the secret. The names of the trees are also changed for the same reason as above. Obviously there would be other varieties of trees on a tropical treasure island.")

Do you see the connection between the triangle theorem proved earlier and the buried treasure problem?

See if you can find a solution to the problem using complex numbers.

There are, of course, “pure geometric” solutions to the treasure problem. We close this article with a diagram that may suggest a solution to you. But we shall give no explanation; you must find that for yourself!

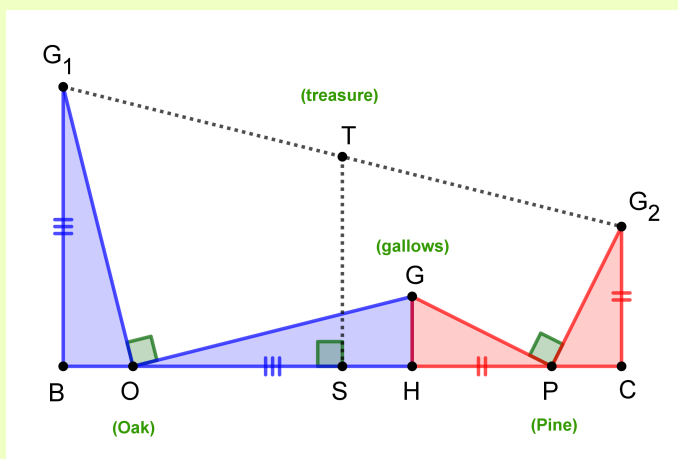


Figure 34: Solution to the buried treasure problem. Adapted from [2]

Further reading

- [1] Daniel Scher, “A Dynamic Approach to Finding Pirate Treasure” from <https://www.sineofthetimes.org/a-dynamic-approach-to-finding-pirate-treasure/>

- [2] Daniel Scher, “Dynamic Visualization and Proof: A New Approach to a Classic Problem”, *The Mathematics Teacher*, Vol. 96, No. 6 (September 2003), pp. 394-398 (5 pages), from

<https://www.jstor.org/stable/20871362>

- [3] Michael de Villiers, “Pirate Treasure Hunt and a Generalization” from

<https://dynamicmathematicslearning.com/treasure-hunt.html>

In [3] the author (Michael de Villiers) considers an interesting generalization of the problem in which the turning angles are not the same but are allowed to vary so long as the sum is always 180° . So, the instruction given is this:

Start from the gallows (G) and walk to the oak tree (O). Turn right through an angle α and move forward by a length GO, to point G_1 . Next, walk from G to the pine tree (P), turn left through an angle $\beta = 180^\circ - \alpha$ and move forward by a length GP, to point G_2 . Bury the treasure at the midpoint of G_1G_2 .

We invite the reader to explore this variation of the problem.

Memorable Sums – 1

Say we wish to compute the following infinite sum:

$$\frac{1}{10} + \frac{1}{100} + \frac{1}{1000} + \frac{1}{10000} + \frac{1}{100000} + \cdots$$

Expressed in another way, we wish to compute the following sum:

$$0.1 + 0.01 + 0.001 + 0.0001 + 0.00001 + \cdots$$

We may prefer to display it in “long addition” format:

$$\begin{array}{r}
 0.1 \qquad \cdots \\
 + \quad 0.01 \qquad \cdots \\
 + \quad 0.001 \qquad \cdots \\
 + \quad 0.0001 \qquad \cdots \\
 + \quad 0.00001 \qquad \cdots \\
 + \quad \cdots \cdots \cdots \cdots \cdots \cdots \\
 \hline
 0.11111 \quad \cdots
 \end{array}$$

The answer is obviously the number $0.11111\dots$, with infinitely many 1’s.

Is this a rational number? That is, can it be expressed as a ratio of non-zero integers?

Yes, surely. We may *recognize* the decimal expansion and see directly that $\frac{1}{9} = 0.1111\dots$

What if our recollection is not good? In that case, let $a = 0.1111\dots$; then,

$$\begin{array}{rcl} 10a & = & 1.1111\dots, \\ a & = & 0.1111\dots, \\ \hline \therefore 9a & = & 1 \quad \text{(by subtraction),} \end{array}$$

giving $a = 1/9$.

A more challenging summation

Now for something more challenging. Let's try to compute the following infinite sum:

$$b = \frac{1}{10} + \frac{2}{100} + \frac{3}{1000} + \frac{4}{10000} + \frac{5}{100000} + \dots$$

Expressed in another way, we wish to compute the following sum:

$$b = 0.1 + 0.02 + 0.003 + 0.0004 + 0.00005 + \dots$$

We may prefer the “long addition” format:

$$\begin{array}{rcl} & 0.1 & \dots \\ + & 0.02 & \dots \\ + & 0.003 & \dots \\ + & 0.0004 & \dots \\ + & 0.00005 & \dots \\ + & \dots\dots\dots & \dots \\ \hline & 0.12345 & \dots \end{array}$$

The answer is the number $b = 0.12345\dots$

It may appear that b is obtained by writing the numbers one after the other consecutively as an infinite decimal:

$$0.12345678910111213\dots,$$

but we would be in error if we supposed this! The above number is *not* b . For, in computing the sum

$$\frac{1}{10} + \frac{2}{10^2} + \frac{3}{10^3} + \frac{4}{10^4} + \cdots + \frac{10}{10^{10}} + \frac{11}{10^{11}} + \cdots,$$

the term $10/10^{10}$ contributes a '1' to the *previous* decimal place; so does the term $11/10^{11}$; and so do all the terms that follow — they all 'collide' with earlier terms. To find the correct decimal expansion, we have to do careful bookkeeping to see what is 'happening' to the carry-overs! If we do this correctly, we're in for a surprise. We find that

$$b = 0.123456790123456790123456790\dots$$

We see a recurring decimal here (with the repeating block 123456790; note the absence of 8; it does not feature anywhere in this decimal expansion). This means that b is a rational number! By applying the standard procedure for finding the fraction corresponding to a recurring decimal, we find that

$$b = \frac{123456790}{999999999},$$

which nicely simplifies after cancelling common factors:

$$b = \frac{10}{81}.$$

Is there a way of working out that $b = 10/81$ without having to get the decimal expansion? There is, and it is quite ingenious.

We have,

$$\begin{aligned}
 b &= \frac{1}{10} + \frac{2}{100} + \frac{3}{1000} + \frac{4}{10000} + \frac{5}{100000} + \cdots, \\
 \therefore \frac{b}{10} &= \frac{1}{100} + \frac{2}{1000} + \frac{3}{10000} + \frac{4}{100000} + \cdots, \\
 \therefore \frac{9b}{10} &= \frac{1}{10} + \frac{1}{100} + \frac{1}{1000} + \frac{1}{10000} + \frac{1}{100000} + \cdots,
 \end{aligned}$$

the last line coming from simple term-by-term subtraction. But we already know what the sum in the last line is; so we get

$$\frac{9b}{10} = \frac{1}{9}, \quad \therefore b = \frac{10}{81},$$

as was to be shown.

One more such summation

Now for something still more challenging. Let's try to compute this infinite sum:

$$c = \frac{1^2}{10} + \frac{2^2}{100} + \frac{3^2}{1000} + \frac{4^2}{10000} + \frac{5^2}{100000} + \cdots.$$

Expressed in another way, we wish to compute the following sum:

$$c = 0.1 + 0.04 + 0.009 + 0.0016 + 0.00025 + \cdots$$

This time we will not try to write the sum in the "long addition" format as it gets rather messy.

Because of the collisions caused by the carry-overs it is not easy to guess the decimal expansion of c . Let's try something similar to what we did last time.

We have,

$$c = \frac{1^2}{10} + \frac{2^2}{100} + \frac{3^2}{1000} + \frac{4^2}{10000} + \frac{5^2}{100000} + \dots$$

$$\therefore \frac{c}{10} = \frac{1^2}{100} + \frac{2^2}{1000} + \frac{3^2}{10000} + \frac{4^2}{100000} + \dots,$$

$$\therefore \frac{9c}{10} = \frac{1}{10} + \frac{3}{100} + \frac{5}{1000} + \frac{7}{10000} + \frac{9}{100000} + \dots,$$

since $2^2 - 1^2 = 3$, $3^2 - 2^2 = 5$, and so on. Now observe that

$$\begin{aligned} & \frac{1}{10} + \frac{3}{100} + \frac{5}{1000} + \frac{7}{10000} + \frac{9}{100000} + \dots \\ &= 2 \left(\frac{1}{10} + \frac{2}{100} + \frac{3}{1000} + \frac{4}{10000} + \dots \right) - \left(\frac{1}{10} + \frac{1}{100} + \frac{1}{1000} + \frac{1}{10000} + \dots \right) \\ &= 2b - a = \frac{20}{81} - \frac{1}{9} = \frac{11}{81}. \end{aligned}$$

Note how we are cleverly making use of our earlier findings.

Hence,

$$\frac{9c}{10} = \frac{11}{81}, \quad \therefore c = \frac{110}{729}.$$

If we had attempted to find the repeating portion by actual calculation, we would be in for a *very* difficult time because for this fraction, the repeating portion is 81 digits long! Here is the 81-digit repeating portion:

150 891 632 373 113 854 595 336 076 817 558 299 039 780

521 262 002 743 484 224 965 706 447 187 928 669 410

Closing comments

Based on the above, would you like to make a guess for the fraction d that corresponds to this sum? —

$$d = \frac{1^3}{10} + \frac{2^3}{100} + \frac{3^3}{1000} + \frac{4^3}{10000} + \frac{5^3}{100000} + \cdots$$

We will reveal the answer in the next issue.

We will also present another way altogether for computing these sums. It will turn out to be more compact than the above, so you may prefer that approach.

A remark about convergence

One more point. We have been freely manipulating infinite series in this article. It turns out that the series we have studied here are of a kind for which such manipulations (and more) can be safely done: rearrangement of terms, re-bracketing of terms, etc. So the results we obtained are valid and accurate. But there are other kinds of infinite series for which familiar manipulations like rearrangement of terms and re-bracketing of terms lead to absurdities (or even to contradictions). We will encounter some of these situations (and learn how to stay away from danger!) in later articles.

More next time!

Proof-Based Grid-Puzzles – 1: The Hopping Horse Puzzle

Article contributed by Rossi D’Souza

Analysing Childhood Puzzles

The use of puzzles as a teaching aid has historically been an inseparable part of mathematics teaching [1], [2]. Mathematical puzzles can therefore be said to have made their presence across ages — ‘ages’, both in terms of historical eras as well as how old we are!

Many of us in our school days would likely have encountered the following puzzle that involves tracing out the diagram shown in Figure 35 without lifting your pencil and without going over any line segment twice.

No matter how much you tried, you would have never succeeded in completing the task.

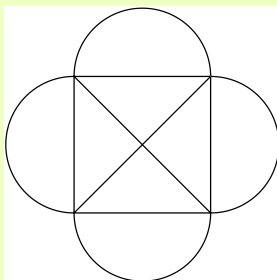


Figure 35: Diagram To Be Traced Without Lifting Your Pencil

You might have also played another puzzle that involves joining points with lines — one that requires tracing four straight line segments through 9 dots arranged as a 3×3 grid (Figure 36), again without lifting your pencil and without going over any line segment twice.

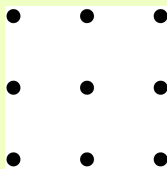


Figure 36: Four lines through 9 points puzzle

Unlike the first puzzle, however, this puzzle can be solved (Figure 37) by realizing that we are allowed to extend a line segment beyond the ‘imaginary 3×3 box’ that we earlier assumed to be part of the conditions or constraints.

In regard to the first puzzle, you may have encountered the proof of its unsolvability only much later in life.

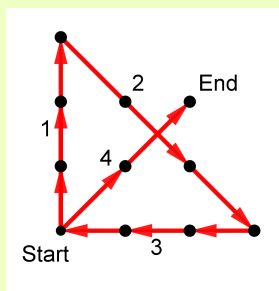


Figure 37: Solution of the Four lines through 9 points puzzle

The proof — based on the work of the 18th century Swiss mathematician Leonard Euler who considered the problem of the Seven Bridges of Königsberg (discussed in detail in the article [3]) — was essentially as follows. Let's consider each vertex of the diagram (as shown in Figure 38) as an island, and regard each line segment (including curved lines) as a bridge. This means we have four islands arranged as $\square ABCD$ which are interconnected by a total of ten bridges.

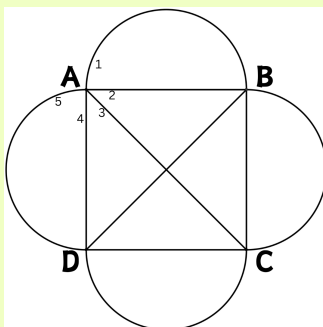


Figure 38: Four Islands arranged as $\square ABCD$ connected by ten bridges

Let's say you begin on island A and travel to island C via a 'bridge,' and then leave island C using an unused bridge and enter island D and then island B, and so on. By the end of your journey, apart from your starting island A and the final island — whichever it might be (it could even be the one where you started the journey) — in the case of every other 'in-between' island, for every bridge that you use to enter the island, you require an unused bridge to exit it. Therefore, other than the starting and final islands, the number of bridges connected to every other bridge *must necessarily be an even number*.

Therefore, a figure that is traceable without lifting your pencil and without retracing a line segment must have *at most two islands* with an odd number of bridges connected to them. The rest of the islands must have an even number of bridges connected to them, on account of their being 'in-between' islands.

However, in our figure we have 4 islands with an odd number of bridges. Therefore, our figure is not traceable under the given conditions.

Even if you consider the point of intersection of AC and BD to be an island with 4 bridges connected to it, it still does not reduce the number of islands with an even number of bridges (in fact, any point on a bridge may be considered an island as well!).

Puzzles that we encountered as children — whether mathematical or logical or brainteasers based on wordplay — were largely in the form of riddles with an answer. However, the question of proving for unsolvability of the first puzzle provides a novel perspective on mathematical puzzles. For Leonard Euler and the mathematical community after him, it meant a new field of mathematics altogether!

Modifying or extending a mathematics problem of a question-answer form into a proof-based puzzle activity offers an insight into mathematics learning. We find that such puzzles bring to the fore many mathematical ideas and get students to appreciate unexpected interconnections.

In this two-part article, I share proof-based grid-puzzle activities that I have developed for teaching and outreach by modifying some old puzzles.

The Hopping Horse Puzzle

The Hopping Horse puzzle is a simplified version of the *The Knight's Tour Problem* which can be described as follows:

A Chess knight is placed at the bottom left square (the 'a1' square in standard chess notation) of a chessboard (or an 8×8 chequerboard). The knight moves according to the standard rules of chess (to complete a 'move' a knight hops over two steps in a given direction and lands on a square in either perpendicular direction). Can you move the knight across the board so that it visits each and every square just once?

The Knight's Tour problem is very challenging, but by using a subset of the chessboard we obtain a less complicated puzzle (in terms of having fewer combinations to try out). Further, it can also be 'modelled' so as to unravel the set of all possible solutions.

So here is the Hopping Horse puzzle.

In the subset of the chessboard shown in Figure 39 (a subset of a 3×5 chequerboard where the middle squares of the edges are excluded), in how many ways can you move a chess knight

across the board so that it visits each square exactly once? (Note: For a chess horse or knight, what matters is the starting and end square, even if the squares skipped over while making a move are occupied, or missing in the case of our puzzle. For example, a move can be made from A to F.)

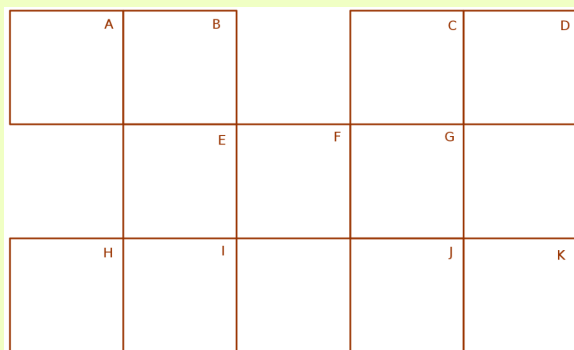


Figure 39: The Hopping Horse Puzzle Board

For example, say the horse begins at square A. From here, it can choose between squares I and F. Say it moves to F and then to H-B-G. From G it can only move to I, from where we find our horse unable to move further — given the condition that it cannot visit a previously visited square (the only squares the horse can move to from I are A and G, both of which were visited earlier) — thus leaving squares C, D, E, J & K unvisited.

However, if the horse starts at H and then hops to B, followed by G-I-A-F-D-J-E-C-K, it will have completed the journey successfully.

By symmetry, another path can be arrived at by starting at another corner square.

This version of the knight's tour problem may appear to involve the use of trial-and-error to arrive at all possible paths. It now needs to be shown that no other paths are possible. For example, we can see that there are at least 4 solutions as there are 4 corners. But does this exhaust all possibilities?

Modelling the problem

To find the set of all possible paths, we model the problem by conceptualizing the 11 squares as dots. We start by drawing a dot and naming it, 'A' (to represent square A) and we connect it with dot I and dot F as the horse can move from A to I or A to F and vice versa in one move. Similarly, we connect I with G, and F with D and K. That is, we connect each pair of dots with a line segment if a horse can move from one square to the other. We then arrive at the diagram in Figure 40.

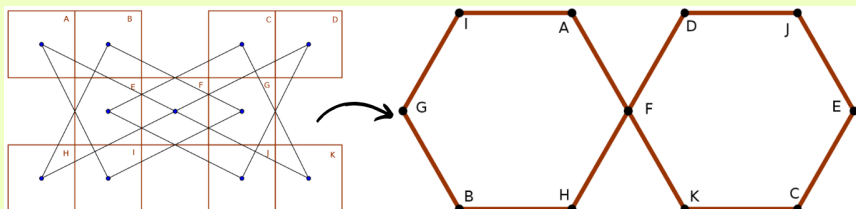


Figure 40: Modeling The Hopping Horse Puzzle

Now by looking at the model of the puzzle — which is made up of two 'loops' — it becomes evident that the only possible solutions are those that begin at any point, travel along the loop in either of two directions — clockwise or anti-clockwise — till you reach F, then travel across the other loop in either of two directions. Consequently, we have 8 possible paths:

1. A-I-G-B-H-F-K-C-E-J-D

2. A-I-G-B-H-F-D-J-E-C-K
3. H-B-G-I-A-F-K-C-E-J-D
4. H-B-G-I-A-F-D-J-E-C-K
5. D-J-E-C-K-F-A-I-G-B-H
6. D-J-E-C-K-F-H-B-G-I-A
7. K-C-E-J-D-F-A-I-G-B-H
8. K-C-E-J-D-F-H-B-G-I-A.

Please note that tracing the horse's path is not the same as tracing a diagram without lifting your pencil. For example, to trace out the figure of 8 or ∞ , one can begin anywhere on the path, which is not the case here.

Discussion

The *Hopping Horse* puzzle is a simplified version of the more complex *Knight's Tour* problem (which some of us who have studied Computer Science in college might have tried to simulate using computer code). But this modification, as I have found, makes it accessible to students without such a background, and I am sure it can serve as a stepping stone into the fields of mathematical modelling and simulation.

I encourage readers to analyse the puzzle for different subsets of the chess-board and share their observations.

Further reading

- [1] Danesi, M. (2004). "The Liar Paradox And The Towers of Hanoi: The Ten Greatest Math Puzzles Of All Time." New Jersey: John Wiley & Sons, Inc.
- [2] Gorev, P. M., N. V. Telegina, L. Z. Karavanova, and S. S. Feshina (2018). "Puzzles as a Didactic Tool for Development of Mathematical Abilities of Junior Schoolchildren in Basic and Additional Mathematical Education". *EURASIA Journal of Mathematics, Science and Technology Education* 14 (10), 178 – 185.
- [3] Samuel, J. "Crossing bridges." *Resonance* 4, 23–31 (1999).
<https://doi.org/10.1007/BF02837149>

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Playing with Solids – 1

Article contributed by Swati Sircar

Three-dimensional geometry and the study of solids moved out of school syllabi historically, to make room for calculus. However, they have now re-entered the textbooks, right from Class 1. The existing NCERT textbooks attempt to build a feeling of a solid geometry through the study of nets and sketches (isometric and oblique) and views (front-top-side). This approach demands a level of intuition that many teachers find difficult, mainly because they rarely get a chance to engage at any depth with solid shapes.

It is worth noting that almost everything we use — textbooks, notebooks, blackboard or screen — is two-dimensional. These are not sufficient to build a good three-dimensional sense. Mathigon-polypad — the virtual manipulative website, <https://mathigon.org/> — does something better, at least for polyhedra. One can create a net of a solid using various polygons and then click on 'Fold Net' to form the solid, which can then be

rotated in any direction, through any angle. However, this is not always accurate. There are some instances where Mathigon does not fold a valid net completely. So, in this article, we will share (a) some easy-to-make modules to create and explore solids, and (b) some potential explorations suitable for Class 6 and beyond.

The modules are polygons surrounded by circular flaps, which are cut halfway (Figure 41). These can be made from boxes. To make a module:

- Draw the polygon on any stiff paper/card
- Find the midpoint of each side of the polygon
- Draw an arc on each side such that the midpoints of the side and the arc are 1 cm apart, i.e., the thickest part of the sector is 1 cm wide
- Cut along the right half each side (of the polygon) till its midpoint
- Crease and fold along the left half of each side

For equilateral triangles and squares, the arcs are essentially the corresponding circumcircle.



(a)



(b)

Figure 41

Modules can be joined along sides with the same length by interlocking the flaps. Once a net is created, it can be folded to form the solid. It is possible to interlock the flaps along each edge. The flaps can remain outside the solid (easier) or inside (more difficult). We recommend that the sides of any polygon be at least 4 cm long — smaller modules are difficult to work with. Let's start with some explorations.

Exploration 1: The cube

The cube is made with six identical or congruent squares. Cuboids are generalisations of a cube, with rectangles replacing some or all of the squares. Here are some questions to explore.

- (a) Can (some or all of) the square faces of a cube be replaced by rhombi?
How many different kinds of solids do we get?
- (b) How many different types of cuboids do we get if we consider a cube as a type for which all dimensions are equal?
- (c) If we replace some faces of a cuboid with parallelograms, do we get some solids? How many different types do we get?

As we try to generalise from the cube to the parallelepiped, we find the following:

A. There are four types of cuboids:

- (a) Cube – all dimensions equal
- (b) Flattened cube – two dimensions equal, and the third shorter than the first two
- (c) Elongated cube – two dimensions equal, and the third longer than the first two

- (d) Brick – all three dimensions different
- B. The cube can be tilted in three ways. So, there are four parallelopipeds with all sides equal:
 - (a) No face tilted
 - (b) Only one pair of opposite faces tilted
 - (c) Two pairs of opposite faces tilted, i.e., only one pair of opposite faces not tilted
 - (d) All faces tilted
- C. The flattened cube and the elongated ones generalise in similar manner to give rise to 6 possible parallelopipeds, while the brick generalises to 8 possible parallelopipeds.

Table 1 (next page) captures all the possibilities.

We explored rhombi and parallelograms with one acute-obtuse linear pair. We can do so with other acute-obtuse linear pairs as well.

Exploration 2: The tetrahedron

We are familiar with regular tetrahedra. Now we shall look for tetrahedra with congruent faces that are: (a) isosceles triangles; (b) scalene triangles.

Tetrahedra with congruent faces It can be proved that tetrahedra with congruent faces can have only acute triangles. The proof is in two parts.

- (i) Any tetrahedron with four congruent faces opens in a way that its net is a bigger triangle similar to each face.
- (ii) Only acute triangles can fold up to form a tetrahedron.

	All three dimensions equal	Two dimensions equal, third shorter	Two dimensions equal, third longer	All three dimensions different
No face tilted	All faces square	Two faces square, rest with smaller area	Two faces square, rest with larger area	No square face
One pair of opposite faces tilted	Four squares and two rhombi	Square faces tilted	Square faces tilted	Smallest faces tilted
		Rectangular faces tilted	Rectangular faces tilted	Medium faces tilted
				Largest faces tilted
Two pairs of opposite faces tilted	Two squares and four rhombi	Square faces not tilted	Square faces not tilted	Smallest faces not tilted
		Rectangular faces not tilted	Rectangular faces not tilted	Medium faces not tilted
				Largest faces not tilted
All faces tilted	All faces rhombi	Two rhombi and four smaller parallelograms	Two rhombi and four larger parallelograms	No face is rhombus

Table 1

The second part of the proof is given in the “Low Floor High Ceiling” article “Triangles to Tetrahedron” in the July 2020 issue of *At Right Angles*.

First part of proof

Consider a tetrahedron $OABC$ for which the faces $\triangle ABC$, $\triangle OAB$, $\triangle OBC$ and $\triangle OCA$ are congruent to one another. Let $\triangle ABC$ be the base, and

suppose that $AB < BC < CA$; see Figure 42. (For simplicity, we assume strict inequalities among the sides.) From $AB < BC < CA$ in $\triangle ABC$, we get $OA, OC < CA$ in $\triangle OCA$ and $AB < OA, OB$ in $\triangle OAB$. It follows that $AB < OA < CA$, and therefore that $OA = BC$. Similarly, $OB = CA$ and $OC = AB$. Therefore,

$$\triangle ABC \cong \triangle OCB \cong \triangle BAO \cong \triangle COA.$$

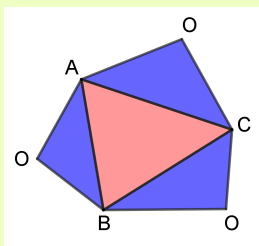


Figure 42

Now if the tetrahedron is cut along OA, OB, OC and flattened out, the cut edges line up and four faces together make a bigger and similar triangle with $\triangle ABC$ at the centre surrounded by three more congruent triangles. (Note that for any triangle, if we connect the midpoints of its sides, we get four congruent smaller triangles, each similar to the original one, with scale factor $1/2$.)

Continuing, there are 8 types of triangles if we consider the sides (scalene, isosceles, equilateral) and the angles (acute, right, obtuse). E.g., the two triangles in a geometry box (the two 'set-squares') are right-scalene and right-isosceles. Similarly, there are obtuse-scalene and obtuse-isosceles

triangles. But there are four types of acute triangles: acute-scalene; isosceles with shorter equal sides; equilateral; and isosceles with longer equal sides.

Now, only acute triangles can fold up to form a tetrahedron as mentioned in the Low Floor High Ceiling article “Triangles to Tetrahedron” in the July 2020 issue of *At Right Angles*. Therefore, if there is a tetrahedron with four obtuse-angled congruent faces, then there is an obtuse triangle that results from cutting and unfolding it. But this is just not possible. We thus arrive at a contradiction! Similarly, we arrive at another contradiction with right-angled triangles. It follows that for a tetrahedra with congruent faces, the faces are acute triangles.

Exploration 3: The octahedron

Surprisingly, this is possible with all 8 types of triangles! Here are the nets (Figure 43). Observe that the acute isosceles triangles can be joined in two different ways resulting in a total of 10 octahedra!

Note that all octahedrons made with scalene triangles have identical vertices opposite to each other, and each such pair is different! If the sides of the triangles are $a \leq b \leq c$, then the three pairs of vertices are as follows: (i) A and A' : where b and c sides meet, and (ii) B and B' : where c and a sides meet, and (iii) C and C' : where a and b sides meet.

On the other hand, for each type of isosceles triangle, at least one octahedron has identical vertices where three equal sides meet the base.

Exploration 4: The icosahedron

This is the most challenging case! It is easy to make this solid with both types of acute isosceles triangles and prove that it is not possible for right

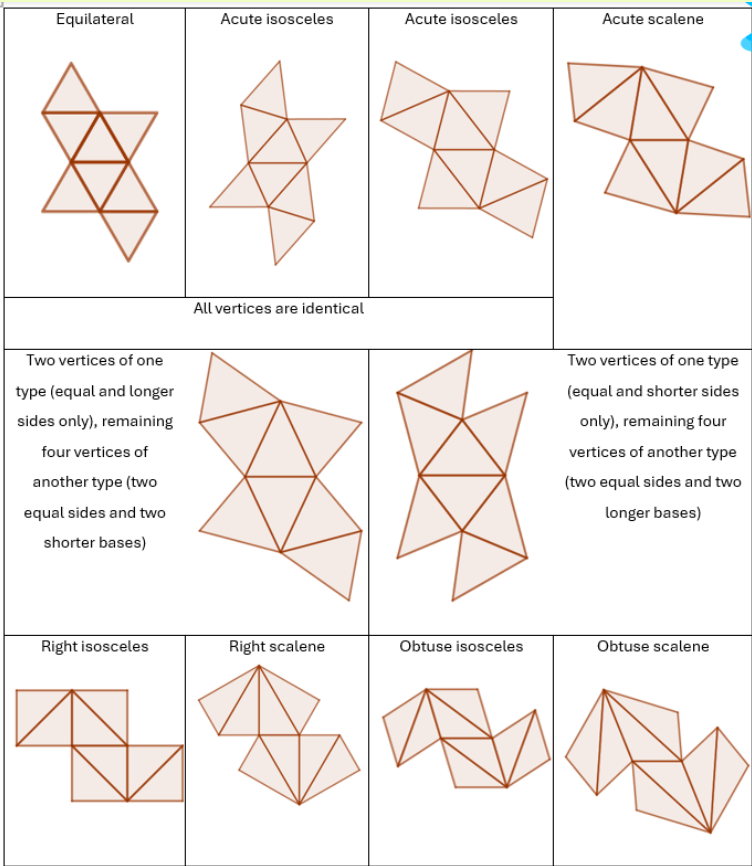


Figure 43

and obtuse isosceles ones. However, scalene triangles are challenging to work with, and we will discuss this possibility in a future article.

Icosahedrons with acute isosceles triangles can be slight modification of

the platonic solid. Here are the two modified nets (Figure 44):



Figure 44

The unequal angles of 5 isosceles triangles converge at each pole – north and south – while the remaining 10 vertices have the angle combination: four equal and one unequal each. This automatically sets an upper limit of $360^\circ / 5 = 72^\circ$ for the unequal angle of the isosceles triangles. Consequently, this arrangement does not work for right triangles and obtuse isosceles triangles. But can there be other possibilities? Will those work for acute isosceles triangles with unequal angle greater than 72° ?

Let us assume that there is an icosahedron whose faces are congruent right isosceles triangles. What are the possible angle combinations at any vertex? There must be at least three acute angles (45°) or at most two right angles to make the angle less than 360° .

So, there are two possibilities: (a) three acute and two right angles; (b) four acute and the remaining right angles.

Now there are two possibilities with (a): (i) the right angles are adjacent, i.e., R-R-A-A-A or (ii) R-A-R-A-A, i.e., there are two acute angles between the right angles. Observe that in both cases: the 5th angle is bounded by equal edges and therefore it must be a right angle, leading to three right

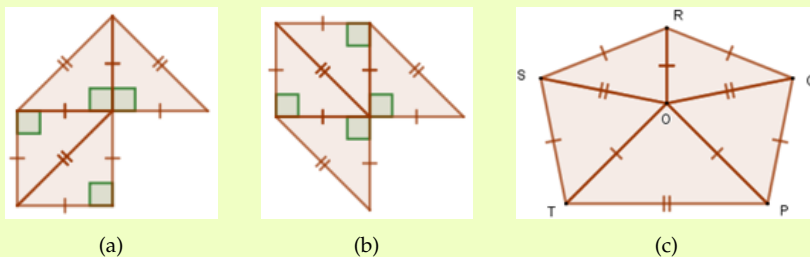


Figure 45

angles and an angle sum of 360° (Figure 45). So (a) is impossible, i.e., no vertex can have two right angles.

If there is a vertex O with just one right angle, it would look like Figure 45(c) where $\angle POT = \angle OPQ = \angle OTS = \angle ORS = \angle ORQ = 90^\circ$, resulting in two right angles at vertex R . Therefore, this too is impossible.

The same argument can be extended for obtuse isosceles triangles and we can prove in a similar manner why that case won't work either. We encourage the reader to work out the details. What about acute isosceles triangles with the unequal angle greater than 72° ? We will discuss that along with the case of scalene triangles in the next article.

Further reading

[1] Math Space, "Triangles to Tetrahedron." *At Right Angles*, Jul 2020, pp.58-66

About the author ...

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Primes in Arithmetic Progression – 1

Have you ever looked for triples of prime numbers in arithmetic progression (AP)? It can be a lot of fun! Here are three such triples: $(3, 5, 7)$, with common difference 2; $(3, 13, 23)$, with common difference 10; and $(3, 23, 43)$, with common difference 20.

What about 4-tuples of primes in AP? For example: $(7, 19, 31, 43)$, with common difference 12. See if you can find a few more. There are lots.

What about 5-tuples? Here are two such: $(5, 11, 17, 23, 29)$, with common difference 6, and $(11, 41, 71, 101, 131)$, with common difference 30. Look for 5-tuples and 6-tuples on your own.

Here is a challenge for you: Find an 8-tuple of prime numbers in AP. Then find a 10-tuple! The list [1] may help you in your search. You'll find that the task gets more and more difficult — but it's still a lot of fun!

After this hard work you will realize how amazing is the result proved by Ben Green and Terence Tao in 2004:

Theorem (Ben Green & Terence Tao). *For any n , however large, there exist n distinct prime numbers in AP.*

That is, it is possible to find sequences of primes in AP of any given length ([3]). But this does not mean that we actually know such sequences! They can be difficult to find. At present the longest such sequence that is known [2] has only 27 primes. The theorem assures us that it should be possible to find a sequence of 1 million primes in AP. But to actually find such a sequence may require superhuman effort!

To conclude, here are a few problems about primes in AP.

1. Let (a, b, c) be a triple of primes in AP with common difference 10. Show that if we append a fourth number d so that a, b, c, d are in AP, then d is **not** a prime number. Show that this conclusion also holds if the common difference is 20.
2. Given a quadruple ('4-tuple') of primes in AP, show that the common difference is a multiple of 6.

Further reading

- [1] Wikipedia, "List of prime numbers" from https://en.wikipedia.org/wiki/List_of_prime_numbers
- [2] Wikipedia, "Primes in arithmetic progression" from https://en.wikipedia.org/wiki/Primes_in_arithmetic_progression
- [3] Wikipedia, "Green–Tao theorem" from https://en.wikipedia.org/wiki/Green%E2%80%93Tao_theorem

On Solar Eclipses – 2

In the July 2024 issue we shared a few quotes about total solar eclipses:

- Although total solar eclipses occur in the same location only every 375 years or so, they are not rare. In fact, they happen on our planet about once every 18 months, on average. [Source: *Live Science*, <http://tinyurl.com/2mtudjym>]
- In the next decade, seven total solar eclipses will occur on Earth, plunging countries as diverse as Australia, Egypt, Spain and Sudan under the ... shadow of the moon. The first ... will be on Aug. 12, 2026, and the last ... on March 20, 2034, with ... durations of totality ranging from 1 minute, 8 seconds to 6 minutes, 23 seconds. [Source: *Live Science*, <https://tinyurl.com/2mtudjym>]
- The longest possible duration of a total solar eclipse is 7 minutes and 32 seconds. The longest annular solar eclipse of the 21st century took place on January 15, 2010, with a duration of 11 minutes and 7.8 seconds. The maximum possible duration is 12 minutes and 29 seconds. [Source: Wikipedia, https://en.wikipedia.org/wiki/List_of_solar_eclipses_in_the_21st_century]

We added that to make such precise statements, there must be a lot of mathematics (geometry, algebra, trigonometry) being used!

We will attempt to justify these statements in this article and a follow-up article.

Geometry of a solar eclipse

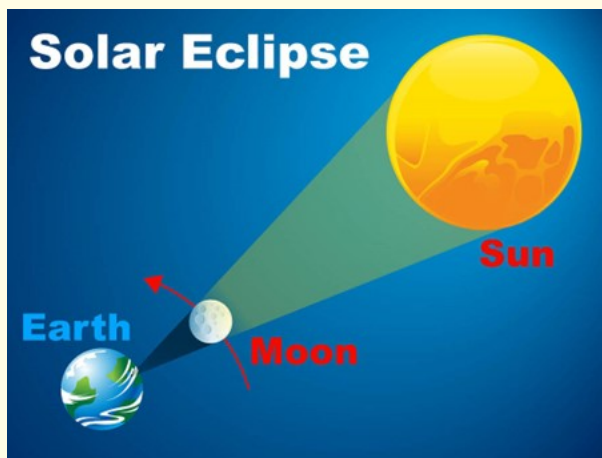


Figure 46: Schematic representation of a solar eclipse. Note that the picture is not drawn to scale! Source: <https://www.nasa.gov/wp-content/uploads/2023/03/solar-eclipse-diagram.jpg>

During an eclipse what we see is just a shadow. It could be the shadow of the Moon falling on the Earth; that's what happens in a solar eclipse (see Figure 46). Or it could be the shadow of the Earth falling on the Moon; that's what happens in a lunar eclipse (see Figure 47). In both cases the

source of light is the Sun. For either type of eclipse to happen, the Sun, the Earth and the Moon must lie (roughly) in a straight line.

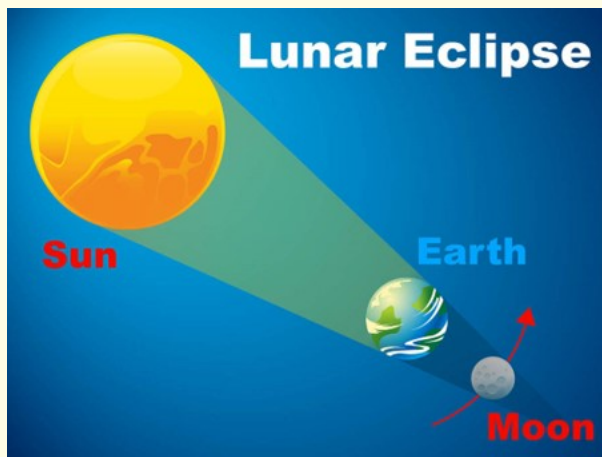


Figure 47: Schematic representation of a lunar eclipse. Here too, the picture is not drawn to scale. Source: https://www.nasa.gov/wp-content/uploads/2023/03/lunar-eclipse-diagram_0.jpg

From the figures, a major difference between the two types of eclipses emerges: a lunar eclipse will be visible to anyone located on the “night side” of the Earth, whereas a solar eclipse will be visible only to those located in the central shadow cast by the Moon.

Again, a lunar eclipse can take place only during the Full Moon phase (when the portion of the Moon that is lit up by the Sun is fully visible to us on the Earth), whereas a solar eclipse can take place only during the New Moon phase (when the portion of the Moon that is lit up by the Sun faces away from us on the Earth and is therefore not visible to us at all).

Since the Moon goes around the Earth once every 27.3 days (effectively 29.5 days if you take into account the Earth's movement in its orbit around the Sun, which is taking place at the same time), it would seem that a lunar eclipse should occur once every 29.5 days, and a solar eclipse too should occur once every 29.5 days. But this is certainly not the case. Both kinds of eclipses occur far less often than that. But why?

The reason is that *the orbit of the Moon is not aligned with the orbit of the Earth around the sun*; it is tilted at an angle of 5.14° relative to the Earth's orbit.

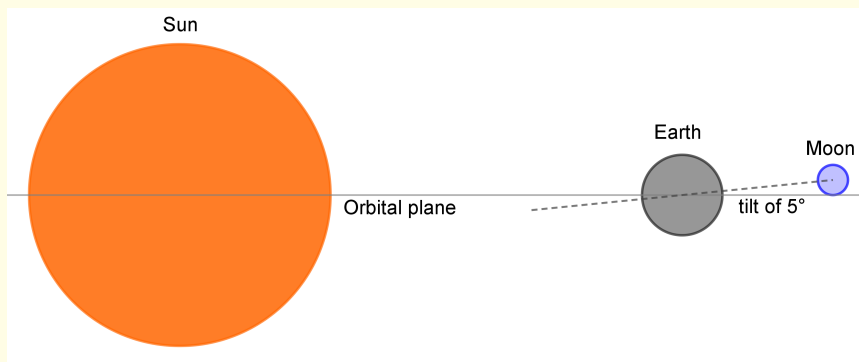


Figure 48: Schematic representation of the Sun, Earth and Moon, seen from the 'side.' Note the 5° tilt of the Moon's orbit relative to the Earth's orbit. (The picture is not drawn to scale.)

This means that for any possibility of an eclipse occurring at the time of the Full Moon or the New Moon, *the Moon must lie in the orbital plane of the Earth* at the time of the Full Moon or New Moon.

But this will happen only occasionally. Most often it will be the case that at

Full Moon or New Moon, the Moon will lie either ‘above’ the orbital plane of the Earth or ‘below’ it.

Given what we have noted above, let us try to estimate how often eclipses will happen. We must remember that the moon is not a point mass; it has a (non-zero) diameter! We will have an eclipse *if the disk of the moon happens to overlap with the orbital plane of the Earth at the time of Full Moon or New Moon*. To proceed with the analysis we now compute a few angles. The diameter of the moon is roughly 3475 km, and the average distance of the moon from the Earth is roughly 384400 km.

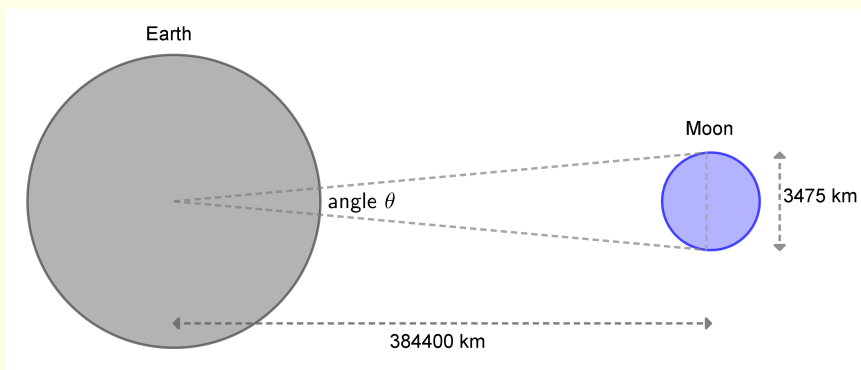


Figure 49: Schematic representation of the Earth and Moon. (This picture, like the others in this article, is not drawn to scale.)

We estimate θ in Figure 49 using the formula “arc length equals $r\theta$ ” (with θ in radians):

$$384400 \cdot \theta = 3475, \quad \therefore \theta = \frac{3475}{384400} \text{ radians} = \frac{3475}{384400} \times \frac{180^\circ}{\pi} = 0.52^\circ.$$

So the angle subtended by the Moon's disk at a point on the Earth's surface is roughly $1/2^\circ$. Note two approximations involved: (a) the distance from the Earth to the Moon is not constant; (b) the Moon moves in an elliptical orbit around the Earth, so there is a minimum distance and a maximum distance, and the angle subtended by the Moon at any given location on the earth varies slightly.

The figure is approximate for other reasons too (e.g., we are treating the figure for the Moon's diameter as an arc length). However, these errors do not amount too much; we do not need to worry about them.

There is something remarkable about the figure we have obtained for the angle subtended by the Moon at the Earth: the angle subtended by the sun at the earth is almost exactly the same!

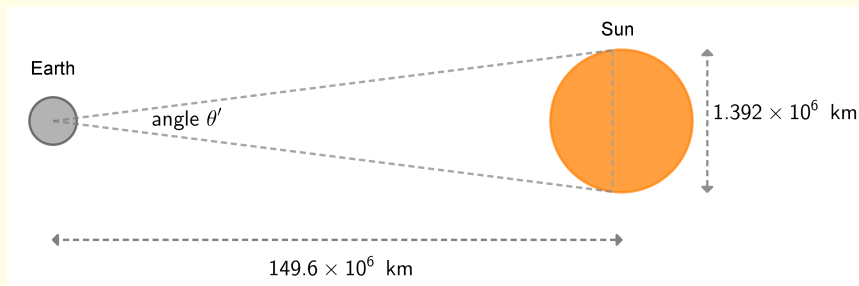


Figure 50: Schematic representation of the Sun and Earth. (This picture, like the others in this article, is not drawn to scale.)

In Figure 50 we have:

$$\theta' = \frac{1.392 \times 10^6}{149.6 \times 10^6} \text{ radians} = \frac{1.392 \times 10^6}{149.6 \times 10^6} \times \frac{180^\circ}{\pi} = 0.53^\circ.$$

Note the closeness of the two figures: 0.52° and 0.53° respectively.

What it implies is that during a total solar eclipse, the disk of the Moon can *exactly* cover the disk of the Sun!

Note that we said “can exactly cover.” As we said earlier, these angles vary by small amounts as the distances are not constants. It may happen during a solar eclipse that the angle subtended by the Moon is *smaller* than that subtended by the Sun. In such a situation the Moon will not fully cover the Sun, and we will get an *annular solar eclipse* instead of a total solar eclipse.



Figure 51: Annular eclipse at sunset in eastern New Mexico, USA. Note the “ring of fire.” Source: https://commons.wikimedia.org/wiki/File:Annular_eclipse_%22ring_of_fire%22.jpg

The fact that these two angles are almost equal to each another may be called a ‘cosmic coincidence’! It will not stay like that forever. This is so

because the Moon is gradually moving further away from the Earth. This in turn happens because the Earth-Moon system is gradually losing energy through tidal friction. Another important consequence of this is that the Earth's rotational speed is gradually reducing. One hundred million years ago (during the time of the dinosaurs), the days were shorter and the moon closer than it is now; it would have looked larger than it is now. One hundred million years from now, the days will be longer than now, and the moon will look smaller!

Back to eclipse computations

Let's see if we can predict the *frequency* of solar eclipses. We know that:

- (a) A solar eclipse can only happen during the New Moon.
- (b) The Moon and Sun both subtend an angle of roughly 0.5° at any point on the Earth's surface.
- (c) The Moon's orbit is tilted at an angle of roughly 5° relative to the Earth's orbital plane.
- (d) A solar eclipse will happen during the New Moon only if the Moon's disk overlaps with the Sun's disk (as seen from the Earth). See Figure 52.

From the above, we conclude that for a solar eclipse to happen, the Moon's centre must lie in a band of width $(0.25 + 0.5 + 0.25)^\circ = 1^\circ$.

As a first approximation, let us argue purely on the basis of proportionality. Since the favourable band has width 1° and the total band of variation has width $(5 + 5)^\circ = 10^\circ$, we conclude that a solar eclipse will occur approximately once every 10 New Moons.

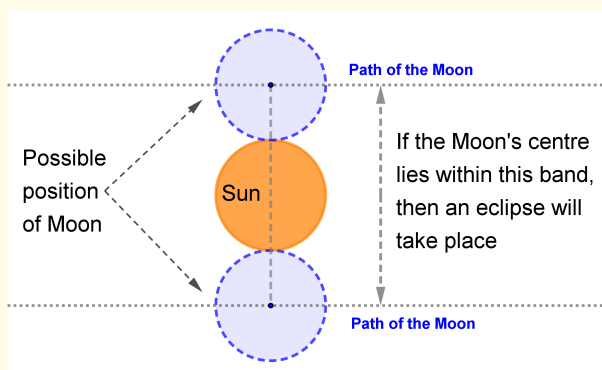


Figure 52: Schematic depiction of the Sun and Moon during the New Moon phase. The Moon is moving from right to left. The 'favourable' situation (in which a solar eclipse occurs) is when the Moon's centre lies within 0.25° of the centre of the Sun. In the 'unfavourable' situation, the Moon passes by the Sun too high or too low, missing the Sun entirely.

The estimate given by this simple-minded reasoning is not too far from the actual figure! The available data suggest that solar eclipses happen "about once every 18 months, on average." Our (rather crude) estimate is "once every 10 months." This may sound a bit 'off' but what should be noted is that the *order of magnitude* of our answer matches that of the actual figure. Considering how simple-minded our reasoning was, this is actually quite impressive.

But is it possible to get a better estimate?

It is, but we must give up the idea of proportional reasoning. A much more careful analysis requires the use of inverse trigonometric functions: we

find that we must compute the quantity

$$2 \times \left(90 - \arccos \frac{0.5}{5} \right)^\circ \approx 11.5^\circ.$$

This tells us that the favourable band has a width of 11.5° out of a total of 180° . Since $11.5/180 \approx 1/16$, we conclude that solar eclipses will happen once every 16 New Moons, or once every 16 months, roughly.

This is a much better estimate! See if you can work out the logic of this on your own.

Note that this analysis only gives us an estimate of how often solar eclipses will happen *somewhere* on the Earth. It says nothing about where the eclipse might be visible and how often solar eclipses will take place at any particular location.

In the continuation of this article we will see how by using elementary geometry, we are able to estimate the duration of a solar eclipse. More in the next issue!

Move Over, Mathematicians, Here Comes AlphaProof

Adapted from: <https://www.nytimes.com/2024/07/25/science/ai-math-alphaproof-deepmind.html> (Article from the New York Times, 25 July 2024). Author: Siobhan Roberts

At Google DeepMind's headquarters in London, a ceremonial gong marks significant achievements in artificial intelligence. This tradition continued with a new breakthrough in mathematical reasoning, as an AI system matched the performance of a silver medallist at the latest International Mathematical Olympiad (IMO 2024).

The achievement is seen as a transformative change in AI's ability to handle complex mathematical problems.

DeepMind's AI models, including AlphaGeometry and a new model called AlphaProof, tackled the Olympiad's challenging problems. The AI solved four out of six problems, earning a score of 28 points, equivalent to a silver

medallist's performance. The system demonstrated proficiency in algebra, geometry, and number theory but struggled with combinatorics.

The AI's performance was evaluated by experts, including Fields medallist Timothy Gowers, who expressed surprise at the AI's success. The system's ability to find "magic keys" to solve problems impressed the judges, marking a step forward in AI-driven mathematical reasoning.

DeepMind's approach involved both informal and formal reasoning systems. The informal system, powered by Google's large language model Gemini, was adept at pattern recognition and creative problem-solving, though it occasionally generated incorrect information. The formal reasoning system used Lean, a theorem prover that ensures logical accuracy.

Reinforcement learning, a key component of previous AI successes like AlphaGo and AlphaZero, was also employed. This approach allows the AI to learn autonomously, potentially surpassing human capabilities over time.

The DeepMind team sees this breakthrough as a milestone in AI's role in mathematics, moving towards artificial general intelligence (AGI). By solving IMO problems, AI systems could eventually become valuable research tools, enhancing accessibility and innovation in mathematics.

While AI's advancement raises questions about the future role of human mathematicians, experts believe that AI will augment rather than replace human efforts. The potential for AI to introduce novel ideas and accelerate research is seen as beneficial, offering new opportunities for collaboration and exploration in the field of mathematics.

See below for the statement of Problem 4 of the IMO which AlphaProof solved.

Problem 4 from IMO 2024

Let ABC be a triangle with $AB < AC < BC$. Let the incentre and incircle of triangle ABC be I and ω , respectively. Let X be the point on line BC different from C such that the line through X parallel to AC is tangent to ω . Similarly, let Y be the point on line BC different from B such that the line through Y parallel to AB is tangent to ω . Let AI intersect the circumcircle of triangle ABC again at $P \neq A$. Let K and L be the midpoints of AC and AB , respectively. Prove that $\angle KIL + \angle YPX = 180^\circ$.

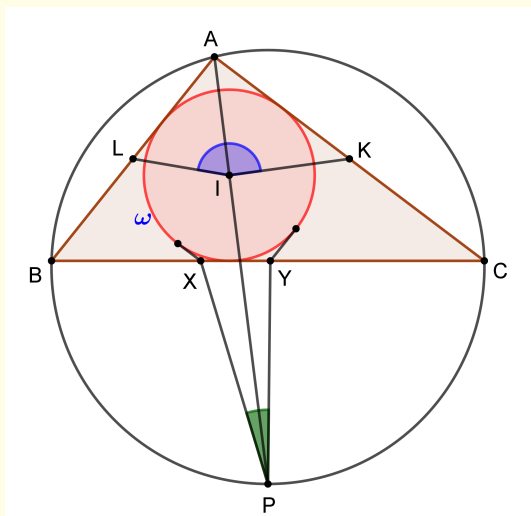


Figure 53: This is Problem 4 of IMO 2024. AlphaGeometry 2 solved it in 19 seconds, after receiving the formalized statement of the problem in AlphaGeometry's own language

Hidden figures: giving history's most overlooked mathematicians their due

Adapted from: <https://www.theguardian.com/books/article/2024/jul/29/secret-lives-numbers-history-mathematicians-overlooked>. Author: David Smith

A new book looks at the contributions made by women and people of colour, whose stories have often been forgotten

The conventional narrative of mathematical history is predominantly a tale of white male geniuses. From Pythagoras to Newton, the spotlight has been narrowly focused. However, *The Secret Lives of Numbers* by Kate Kitagawa and Timothy Revell offers a radical re-evaluation, revealing a far more diverse and interconnected history of mathematical thought.

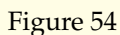
The authors contend that the origins of many fundamental mathematical concepts are often misattributed. Calculus, for instance, is typically credited to Newton and Leibniz, but the book posits that its roots lie in 14th-century India, with the work of Madhava of Sangamagrama. This revelation underscores the book's central argument: that mathematical knowledge has been a global endeavour, shaped by contributions from cultures worldwide.

To illustrate this point, the book profiles a range of extraordinary mathematicians who have been overlooked or marginalized. Hypatia, a brilliant Alexandrian philosopher and mathematician, was tragically murdered for her intellectual pursuits. Sophie Kowalevski, defying societal expectations, became a pioneering female mathematician, making significant strides in her field.

Beyond Europe, the book explores the rich mathematical traditions of the Islamic world, exemplified by the House of Wisdom in Baghdad. This centre of learning, active in the 8th century, was a crucible of intellectual exchange, where scholars from various cultures collaborated to translate, study, and expand upon existing knowledge.

The Secret Lives of Numbers also highlights the contributions of Chinese mathematicians, such as Ban Zhao, an early female mathematician and astronomer. The book emphasizes the interconnectedness of mathematical ideas, demonstrating how concepts developed in one region often influenced work in others.

By challenging the Eurocentric bias in mathematical history, the authors invite readers to reconsider their understanding of the discipline. They argue that mathematics is a collaborative, global enterprise, shaped by countless individuals from diverse backgrounds.



The Secret Lives of Numbers is a call to reimagine the history of mathematics as a complex, multifaceted narrative. By recognizing the contributions of mathematicians from different cultures and backgrounds, we gain a deeper and more accurate understanding of this fundamental human endeavour.

Why ‘sling action’ bowling deceives so many batsmen

Adapted from: <https://www.newscientist.com/article/2443871-why-sling-action-bowling-deceives-so-many-batters-in-cricket/> From: *New Scientist*, 15 August 2024.
Author: James Woodford

Physicists have unveiled the secrets behind the highly effective horizontal “sling-action” bowling technique in cricket. The sling-action method, popularized by Sri Lankan cricketer Lasith Malinga (who was the first bowler to take 100 wickets across all three formats of international cricket), involves extending the arm at about 90 degrees from the body, releasing the ball almost parallel to the ground.



Figure 55: Lasith Malinga in action

Researchers at Amity University Dubai used a wind tunnel to study the air flow around a spinning cricket ball, rotating at up to 2500 revolutions per minute, released from this lower horizontal position. They discovered that after release, the ball's seam spins almost parallel to the ground throughout its flight. This creates a situation where one half of the ball spins opposite to the airflow, while the other half spins towards it.

(This is the case for the bowling of any fast bowler but in this case the effect is greatly enhanced because the seam of the ball is almost parallel to the ground.)

This spinning motion generates a pressure difference, causing the ball to drift laterally during its flight. The magnitude of this drift depends on the size of low-pressure zones near the ball and how they change due to spin.

Higher bowling speed and spin rate increase the lateral drift, making it more challenging for batsmen to predict the ball's trajectory.

The researchers found that horizontal-arm bowlers unconsciously employ the *Magnus effect* ([1]) — a physics concept where spinning motion combines with airflow to produce sideways movement. This effect is also seen in other sports, such as football free kicks (the “banana kick”) and table tennis shots.

While the study explains some aspects of horizontal bowling's effectiveness, experts believe other forces are also at play. Simon Watkins from RMIT University in Melbourne suggests that very small adjustments, such as where a bowler holds and releases the seam, can dramatically alter the ball's direction.

The research highlights how cricketers and other athletes have intuitively exploited complex aerodynamic principles for generations, often without fully understanding the underlying physics. This study provides valuable insights into the science behind the art of bowling, potentially helping players and coaches refine their techniques and strategies in the future.

Further reading

[1] Wikipedia, “Magnus effect” from

https://en.wikipedia.org/wiki/Magnus_effect

[2] Encyclopedia Britannica, “Magnus effect” from

<https://www.britannica.com/science/Magnus-effect>

Physics Nobel scooped by machine-learning pioneers

Adapted from: <https://www.nature.com/articles/d41586-024-03213-8>. Author: Davide Castelvechi

Two prominent researchers, John Hopfield and Geoffrey Hinton, have been awarded the 2024 Nobel Prize in Physics for their groundbreaking contributions to the field of artificial intelligence (AI), specifically in developing artificial neural networks. The announcement was made by the Royal Swedish Academy of Sciences on October 8, 2024, and the prize amount is 11 million Swedish kronor (approximately US\$1 million), which will be shared equally between the two laureates.

Contributions to AI

John Hopfield, a theoretical biologist at Princeton University, introduced the concept of associative memory in 1982. This model allowed for the

storage and reconstruction of patterns by simulating connections between virtual neurons as physical forces. By representing memories as low-energy states, Hopfield's network could recall information based on related cues, mimicking human memory processes. His work laid the foundation for understanding complex data patterns and has been instrumental in various scientific fields, including physics and neuroscience.

Geoffrey Hinton, often referred to as the "Godfather of AI," further developed Hopfield's ideas by incorporating principles from statistical physics into a layered version of the Hopfield network. Hinton's innovations enabled machines to recognize and classify images autonomously, marking a significant advancement over traditional computation methods that relied on step-by-step calculations. His techniques have become foundational in training modern AI systems, including large language models and machine-learning algorithms.

Impact on Science and Society

The Nobel committee highlighted that artificial neural networks have revolutionized research across diverse areas such as particle physics, material science, and astrophysics. These networks are now embedded in everyday applications like facial recognition and language translation. Ellen Moons, chair of the Nobel committee, emphasized that while these technologies offer substantial benefits, they also pose ethical concerns that require careful management to ensure they are used safely and responsibly.

Hinton expressed his astonishment at receiving the award, describing it as a "bolt from the blue." He noted that advancements in machine learning could have impacts comparable to those of the Industrial Revolution but raised alarms about potential risks associated with AI surpassing hu-

man intelligence. He underscored his belief that these technologies could reshape society in profound ways.

Recognition of Interdisciplinary Research

Both laureates have previously received prestigious awards for their contributions to science. Hinton was awarded the Alan Turing Award in 2018, while Hopfield has received several accolades in physics, including the Dirac Medal. Their recognition by the Nobel committee reflects a growing appreciation for interdisciplinary approaches that blend concepts from physics, biology, and computer science.

Experts in the field have praised their work for bridging gaps between disciplines. For instance, physicist Karl Jansen remarked on how Hopfield's motivation stemmed from physics to understand complex systems like phase transitions. Lenka Zdeborová from EPFL noted that recognizing physics ideas in AI highlights their universal applicability across various domains.

Broader Implications for Neuroscience

The implications of Hopfield's and Hinton's work extend into neuroscience as well. May-Britt Moser, a neuroscientist and Nobel laureate herself, pointed out how versions of Hopfield's network models have aided researchers in studying neuronal interactions related to memory and navigation. The ability to analyze vast amounts of data using machine learning tools has transformed our understanding of brain functions.

Eve Marder from Brandeis University emphasized that machine learning is reshaping data analysis in neuroscience, allowing scientists to explore ques-

tions previously deemed impossible. The integration of these technologies into research methodologies continues to enhance our comprehension of how neural circuits operate.

Conclusion

The awarding of the Nobel Prize in Physics to John Hopfield and Geoffrey Hinton underscores their pivotal roles in advancing artificial intelligence through innovative applications of physics principles. Their work not only revolutionizes technology but also raises critical questions about ethics and societal impact as AI continues to evolve. As we navigate this new frontier, their contributions serve as a reminder of the profound intersections between diverse scientific fields and their collective potential to shape our future.

Further reading

- [1] The Royal Swedish Academy of Sciences, “The Nobel Prize in Physics 2024” from <https://www.nobelprize.org/prizes/physics/2024/press-release/>
- [2] Georgina Rannard & Graham Fraser (BBC), “‘Godfather of AI’ shares Nobel Physics Prize” from <https://www.bbc.com/news/articles/c62r02z75jyo>

Capturing rogue planets from 3.8 light years away

Adapted from: <https://www.newscientist.com/article/2441472-the-sun-could-capture-rogue-planets-from-3-8-light-years-away/>

Article from *New Scientist*, 29 July 2024. Author: Jonathan O'Callaghan

This article discusses a new mathematical model that suggests the Sun's gravitational influence may extend much farther than previously thought, potentially redefining the boundaries of our solar system.

Researchers have proposed a mathematical model indicating that the sun's gravitational pull could capture objects from as far as 3.8 light years away. This distance is significantly beyond the Oort cloud, which has traditionally been considered the outer edge of the solar system at just over 1 light year from the Sun.

The model identifies Lagrange points between the Sun and the centre of

the galaxy, similar to those found within our solar system. These points of gravitational balance could create entry points for objects to be pulled into unusual orbits around the Sun. The orbits of these captured objects would form complex, fractal-like patterns resembling Mandelbrot sets.

This extended region of influence could potentially trap various celestial bodies, including interstellar comets, asteroids, and even rogue planets.

Objects caught in these orbits would likely remain there indefinitely unless disturbed by other gravitational forces, such as those from nearby stars like Proxima Centauri or Barnard's star.

The researchers suggest that this region is scientifically significant and worth exploring. Future astronomical surveys, such as those planned for the Vera Rubin Observatory, could search for objects captured in these distant orbits.

If confirmed, this phenomenon could lead to a redefinition of the solar system's boundary, extending it to 3.81 light years from the sun.

The mathematical work behind this model is considered novel, particularly in its application of fractal patterns to orbits in chaotic, restricted three-body environments. However, some scientists question whether this region could actually capture planets, given that it hasn't captured stars.

This research touches on the complex three-body problem in astronomy, which involves calculating the orbit of an object under the influence of two significant gravitational sources – in this case, the Sun and the galactic centre.

If proven correct, this model could have significant implications for our understanding of the solar system's extent and its interactions with interstellar objects.

It also opens up new avenues for astronomical observation and theoretical research into the dynamics of our cosmic neighbourhood.

More on rogue planets, from Wikipedia

A rogue planet, also termed a free-floating planet (FFP) or an isolated planetary-mass object (iPMO), is an interstellar object of planetary mass which is not gravitationally bound to any star or brown dwarf.

Rogue planets may originate from planetary systems in which they are formed and later ejected, or they can also form on their own, outside a planetary system. The Milky Way alone may have billions to trillions of rogue planets, a range the upcoming Nancy Grace Roman Space Telescope will likely be able to narrow.

Some planetary-mass objects may have formed in a similar way to stars, and the International Astronomical Union has proposed that such objects be called *sub-brown dwarfs*.

See https://en.wikipedia.org/wiki/Rogue_planet for more about this.

Why are there so many rogue planets?

Adapted from: <https://www.newscientist.com/article/mg26234921-000-why-are-there-so-many-rogue-planets-and-what-do-they-look-like/>. Article from *New Scientist*, 20 May 2024. Author: Abigail Beall

Rogue planets are worlds in perpetual darkness, lacking a sun to provide light or mark the passage of time. Recent research suggests they may be surprisingly common, potentially outnumbering stars by up to 20 times. This could mean trillions of rogue planets exist in our galaxy alone.

The existence of these free-floating planets aligns with current planetary formation theories. They are believed to form in several ways: some may begin like stars, with gas clouds collapsing under gravity, while Earth-sized rogues likely formed within solar systems before being ejected. Ejection could occur due to gravitational interactions with passing stars or conflicts

with other planets in the system. The most probable explanation for rogue planets in the Milky Way is that they formed around binary star systems and were subsequently ejected.

Detecting rogue planets is challenging as they can't be seen directly. Since 2012, astronomers have inferred their presence through gravitational microlensing, where the planets bend light from distant stars and galaxies. Most detected rogue planets appear to be around Earth's size.

The conditions on rogue planets are expected to be diverse, ranging from rocky worlds to gas giants and icy planets larger than Neptune. Without a host star for light and heat, these planets likely have frozen surfaces. However, internal heat sources could exist, such as thermal vents powered by the planet's contraction as it cools down, similar to the cryovolcanism observed on Pluto.

Some scientists have proposed that rogue planets with large amounts of hydrogen in their atmospheres could maintain Earth-like surface temperatures due to the greenhouse effect at high pressures. Heat could be generated by radioactive decay in the planet's interior.

Simulations have even suggested that some rogue planets might be potentially habitable. This could occur in liquid oceans beneath icy crusts or on the surface if the planet can maintain a thick hydrogen atmosphere to trap sufficient heat for life.

The concept of rogue planets challenges our traditional understanding of planetary systems and opens up new possibilities for the existence and distribution of worlds in our galaxy. As our detection methods improve, we may gain more insights into these wandering worlds and their potential for harbouring life in the vast, cold expanses of interstellar space.

India's pioneering mission bolsters idea that Moon's surface was molten

Adapted from: <https://www.nature.com/articles/d41586-024-02695-w> From *New Scientist*, 21 August 2024, Author: Michael Marshall

India's Chandrayaan-3 mission has made significant strides in lunar exploration by obtaining the first measurements of the Moon's soil composition near its south pole. The mission's Vikram lander successfully touched down on the lunar surface on August 23, 2023, deploying the Pragyan rover, which conducted a 10-day study collecting various data, including temperature and seismological readings.

Among its key tasks, Pragyan focused on analysing the chemical composition of the regolith — the fine material that covers much of the lunar

surface — using an alpha-particle X-ray spectrometer (APXS).

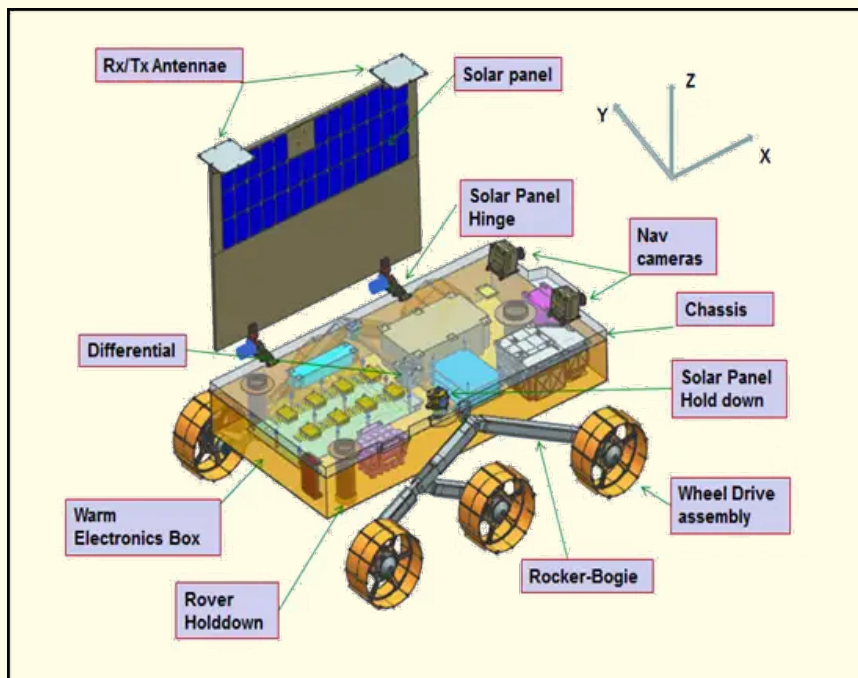


Figure 56: The Chandrayaan-3 Pragyan rover. Source: https://commons.wikimedia.org/wiki/File:Chandrayaan-3_Pragyan_rover.png

The spectrometer was deployed at 23 different locations, and the collected data revealed that the regolith contained primarily Ferroan Anorthosite, a mineral commonly found on the Moon. This finding aligns with previous orbital data, confirming earlier theories about the Moon's composition.

However, Chandrayaan-3's significance lies in its samples being the first to



Figure 57: The Moon probably originated from material scattered into space when a large impactor struck the newly formed Earth. Credit: David Gannon/AFP via Getty (from *New Scientist*)

come from the Moon's subpolar region, contrasting with previous missions that explored equatorial and mid-latitude zones. The uniformity in soil composition across these regions supports the theory that the lunar surface was once entirely covered by a molten magma ocean.

This molten state is thought to have occurred shortly after the Moon formed, possibly as a result of a massive impact event involving a Mars-sized body named Theia — a collision with the newly formed Earth.

This collision is believed to have vaporized part of the Earth's surface, sending debris into orbit, which eventually coalesced to form the Moon. The immense energy from this impact would have melted the young Moon's surface, creating a magma ocean. Over time, denser, metal-rich

rocks sank while lighter materials, including anorthosite, rose to form the lunar highlands.

Interestingly, the Chandrayaan-3 samples showed higher levels of magnesium relative to calcium, suggesting that deeper mafic materials, typically found in the Moon's interior, might have mixed into the regolith, potentially due to the formation of the nearby South Pole–Aitken basin. This enormous crater, located 350 km from the lander's site, likely resulted from an impact that excavated deeper layers of the Moon's crust.

However, the mineral composition in the South Pole–Aitken basin, particularly the dominance of pyroxene, does not entirely align with Pragyan's data, indicating that further analysis and perhaps sample return missions will be necessary to fully understand the composition.

The success of Chandrayaan-3 underscores the advancements of India's space program, with future missions potentially aiming to bring lunar samples back to Earth for detailed study.

Bacteria can work as a team to spot prime numbers and vowels

Adapted from: <https://www.newscientist.com/article/2448167-bacteria-can-work-as-a-team-to-spot-prime-numbers-and-vowels/> From *New Scientist*, 27 September 2024.

Author: Matthew Sparkes

Genetically engineered bacteria have been developed to perform computational tasks such as identifying prime numbers, vowels, or calculating the maximum number of pizza slices from straight cuts.

The research, conducted by Sangram Bagh and his team at the Saha Institute of Nuclear Physics in Kolkata, India, shows that these bacteria, derived from *Escherichia coli*, can act as tiny biological computers.

These bacteria, dubbed “bactoneurons,” are modular and can work together to solve problems, much like a multicellular organism.

In classical computers, information is transmitted using binary numbers (0s and 1s) represented by voltages. In contrast, bactoneurons use chemicals to represent binary input. By manipulating the presence or absence of specific chemicals, these bacteria can perform tasks like determining whether a number is prime or a letter is a vowel. The bactoneurons fluoresce in different colours — green or red — based on the result. They can also calculate how many slices a pizza can be cut into using straight lines.

These bactoneurons, only 2 to 5 micrometers long, are equipped with input, output, and their own chemical power supply. They have advantages over traditional computers in size and cost, as they can self-replicate, enabling mass production at low cost. Bagh suggests that these biological computers could surpass classical computers in miniaturization.

Previous biological computer research used single-celled organisms to process data, but this study marks the first demonstration of a programmable biological computer capable of solving multiple problems. The bacteria function similarly to an artificial neural network, the same architecture used in large language models like ChatGPT, although on a much smaller and simpler scale.

Heron's Formula for Area of a Triangle – 2

In the July issue we presented a proof of Heron's beautiful formula for the area (K) of a triangle with sides a, b, c ,

$$K = \sqrt{s(s-a)(s-b)(s-c)}.$$

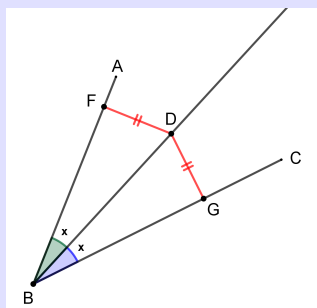
The proof was based on Pythagoras Theorem and was algebraic in nature. It made heavy use of the “difference of two squares” factor formula. Now we present a proof (adapted from [1]) which uses basic trigonometry and some circle geometry. In the January 2025 issue, we shall study a third proof — the one given originally by Heron.

Trigonometric proof

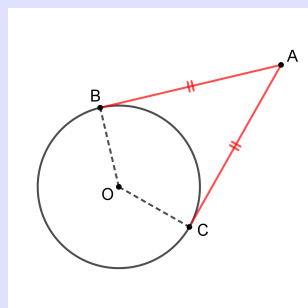
The proof requires some identities and a few basic facts from geometry:

- Each point on the internal bisector of an angle is equidistant from the two arms of the angle.

- From a point outside a circle, two tangents can be drawn to the circle, and they have equal length.



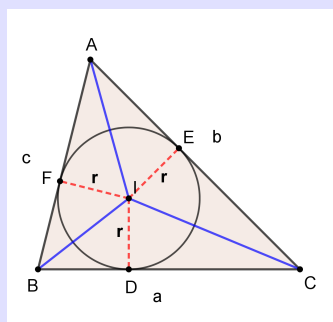
(a)



(b)

Figure 58: Two well-known facts from geometry

Next, we have an important formula connecting K, r, s :



- Area of $\triangle IBC = \frac{1}{2}ar$
- Area of $\triangle ICA = \frac{1}{2}br$
- Area of $\triangle IAB = \frac{1}{2}cr$
- Therefore area of $\triangle ABC$ is $\frac{1}{2}(ar + br + cr) = s \cdot r$
- Hence $K = rs$

Figure 59: An important formula: Area of triangle, $K = rs$

We also have the following identity for the half-angles of a triangle:

Lemma. For any $\triangle ABC$,

$$\tan \frac{A}{2} \tan \frac{B}{2} + \tan \frac{B}{2} \tan \frac{C}{2} + \tan \frac{C}{2} \tan \frac{A}{2} = 1.$$

This follows easily from the following,

$$\frac{A}{2} = 90^\circ - \left(\frac{B}{2} + \frac{C}{2} \right), \quad \tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \cdot \tan y},$$

and $\tan(90^\circ - x) = 1/\tan x$. For,

$$\tan \frac{A}{2} = \frac{1}{\tan(\frac{1}{2}B + \frac{1}{2}C)} = \frac{1 - \tan \frac{1}{2}B \cdot \tan \frac{1}{2}C}{\tan \frac{1}{2}B + \tan \frac{1}{2}C},$$

and cross-multiplication now yields the stated identity. There is also a beautiful proof without words (it is from [1]) which we display below — but without words!

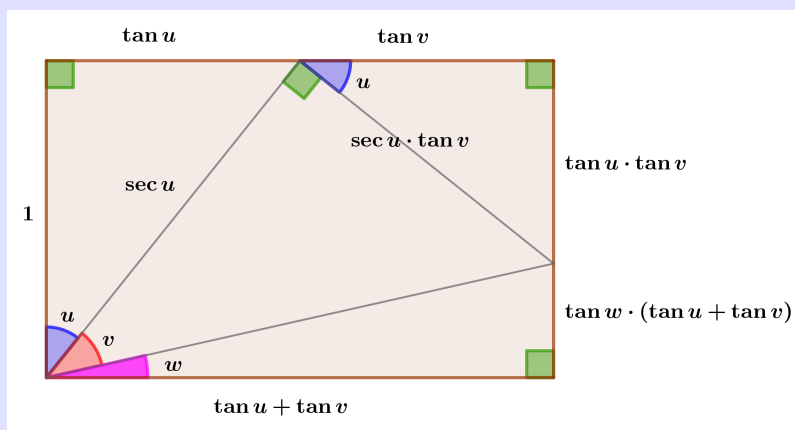


Figure 60: $u + v + w = 90^\circ \implies \tan u \cdot \tan v + \tan v \cdot \tan w + \tan w \cdot \tan u = 1$

All the ingredients have been assembled and we are ready to bake the cake!

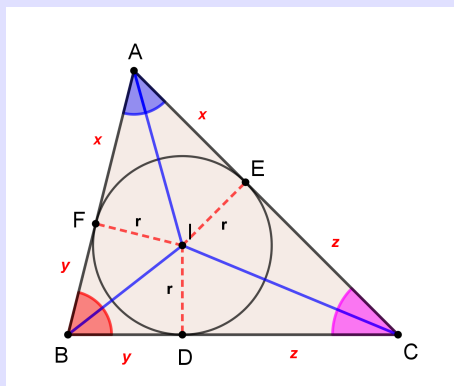


Figure 61

The figure shows $\triangle ABC$ with its incircle and perpendiculars ID , IE and IF drawn from its incentre I to the sides BC , CA and AB . Let $AE = x = AF$, $BD = y = BF$, $CD = z = CE$. We see that

$$a = y + z, \quad b = z + x, \quad c = x + y,$$

and so $a + b + c = 2(x + y + z)$, giving $x + y + z = s$ and,

$$x = s - a, \quad y = s - b, \quad z = s - c.$$

Next, note the presence of the half angles,

$$\angle IAE = \angle IAF = \frac{A}{2}, \quad \angle IBD = \angle IBF = \frac{B}{2}, \quad \angle ICD = \angle ICE = \frac{C}{2}.$$

and,

$$\tan \frac{A}{2} = \frac{r}{x}, \quad \tan \frac{B}{2} = \frac{r}{y}, \quad \tan \frac{C}{2} = \frac{r}{z}.$$

Invoking the tangent identity proved above, we get,

$$\frac{r}{x} \cdot \frac{r}{y} + \frac{r}{y} \cdot \frac{r}{z} + \frac{r}{z} \cdot \frac{r}{x} = 1,$$

$$\therefore r^2 \left(\frac{1}{xy} + \frac{1}{yz} + \frac{1}{zx} \right) = 1.$$

This may be rewritten as

$$r^2 \cdot \left(\frac{x+y+z}{xyz} \right) = 1, \quad \therefore \frac{r^2 s}{xyz} = 1,$$

which yields

$$r^2 s^2 = sxyz.$$

But $K = rs$. Hence

$$K^2 = sxyz.$$

which is just Heron's formula:

$$K^2 = s(s-a)(s-b)(s-c).$$

An elegant proof indeed! ■

Closing note

The concluding part of this article will appear in the January 2025 issue. We will describe Heron's proof and extensions to quadrilaterals.

Further reading

[1] Roger Nelsen, "Heron's Formula via Proof Without Words", Volume 32, No. 4, Sept 2001, *The College Mathematics Journal*, pages 290–293; can

be downloaded here: <http://johnng.inscyber.net/proof-without-words-heron.pdf>

[2] Christy Williams, Crystal Holcomb, and Kayla Gifford, "Heron's Formula for Triangular Area." <https://www.ms.uky.edu/~corso/teaching/math330/heron.pdf>

[3] Wikipedia, "Heron's formula." https://en.wikipedia.org/wiki/Heron's_formula

Article contributed by Shailesh Shirali

e is irrational

The reader may be familiar with the fact that the well-known constant e is an irrational number. Here is an interesting proof of this which we have adapted from [1]. It is different from the proof often quoted in textbooks.

The constant e is generally defined as follows:

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n.$$

(There are many ways in which e can be defined; they are all equivalent to one another.) From the above, we may show that for any x ,

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots.$$

The substitution $x = -1$ yields

$$\frac{1}{e} = \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \cdots.$$

We shall use the last equality to prove the irrationality of $1/e$.

We shall make use of the following result which is true for *any alternating series whose terms are strictly decreasing in absolute value*. Here, ‘alternating’

signifies that the terms are alternately positive and negative. If it also happens that the terms are strictly decreasing in absolute value, then the series converges; i.e., it has a limiting sum. Note that the above series for $1/e$ satisfies both these conditions. Here is the statement of the theorem.

Theorem. *Given an alternating series whose terms are strictly decreasing in absolute value, say*

$$a_1 - a_2 + a_3 - a_4 + \cdots,$$

where $a_1 > a_2 > a_3 > \cdots > 0$, the limiting sum L lies between every pair of consecutive partial sums.

Since the partial sums of the above series are the quantities $a_1, a_1 - a_2, a_1 - a_2 + a_3, a_1 - a_2 + a_3 - a_4, \dots$, the theorem is telling us that the limiting sum L satisfies the following inequalities:

$$a_1 - a_2 < L < a_1,$$

$$a_1 - a_2 < L < a_1 - a_2 + a_3,$$

$$a_1 - a_2 + a_3 - a_4 < L < a_1 - a_2 + a_3,$$

and so on. Justifying these relations is easy. To see why the first relation is true, consider:

$$L = a_1 - (a_2 - a_3) - (a_4 - a_5) - (a_6 - a_7) - \cdots < a_1,$$

since all the bracketed terms are positive; and

$$L = (a_1 - a_2) + (a_3 - a_4) + (a_5 - a_6) + \cdots > a_1 - a_2,$$

since all the bracketed terms are positive. Essentially the same logic works for the other inequalities. Try justifying the next inequality for yourself the same way.

The above series for $1/e$ has two additional features:

- The numerator of each term is ± 1 ; and
- The denominator of each term is a divisor of the denominator of the next term, since $(k-1)!$ is a divisor of $k!$ for all k .

These two facts acting in concert lead to a lovely conclusion.

Let a_n be the partial sum till the term with denominator $n!$,

$$a_n = \frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{(-1)^n}{n!}.$$

Then $1/e$ lies between a_{n-1} and a_n for all n . That is, e lies between these two quantities,

$$a_{n-1} = \frac{1}{2!} - \frac{1}{3!} + \cdots \pm \frac{1}{(n-1)!},$$

$$a_n = \frac{1}{2!} - \frac{1}{3!} + \cdots \pm \frac{1}{(n-1)!} \mp \frac{1}{n!}.$$

Clearly, a_{n-1} can be written as a fraction with denominator $(n-1)!$, and a_n can be written as a fraction with denominator $n!$. Since $(n-1)!$ is a divisor of $n!$, it follows that a_{n-1} too can be written as a fraction with denominator $n!$. So we may write,

$$a_{n-1} = \frac{1}{2!} - \frac{1}{3!} + \cdots \pm \frac{1}{(n-1)!} = \frac{A}{n!},$$

$$\text{and } a_n = \frac{1}{2!} - \frac{1}{3!} + \cdots \pm \frac{1}{(n-1)!} \mp \frac{1}{n!} = \frac{B}{n!},$$

for some positive integers A and B . Studying the expressions for a_{n-1} and a_n carefully, we see that

$$a_n = a_{n-1} \mp \frac{1}{n!}, \quad \therefore \frac{B}{n!} = \frac{A}{n!} \mp \frac{1}{n!}.$$

The two fractions on the far right have the same denominator. Hence,

$$B = A \mp 1,$$

that is, A and B are consecutive integers. This means that $1/e$ lies *strictly* between two fractions with the same denominator ($n!$), whose numerators differ by 1. Moreover, this is true for every positive integer n .

We illustrate this finding with some actual calculation. Here is a list of values of a_n and $n!$ for a few small values of n . The last row gives the values of a_n with denominator $n!$.

n	2	3	4	5	6	7	8
a_n	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{3}{8}$	$\frac{11}{30}$	$\frac{53}{144}$	$\frac{103}{280}$	$\frac{2119}{5760}$
$n!$	2	6	24	120	720	5040	40320
a_n	$\frac{1}{2}$	$\frac{2}{6}$	$\frac{9}{24}$	$\frac{44}{120}$	$\frac{265}{720}$	$\frac{1854}{5040}$	$\frac{14833}{40320}$

Observe the pattern:

- For $n = 3$, we have $\frac{2}{6} < \frac{1}{e} < \frac{3}{6}$.
- For $n = 4$, we have $\frac{8}{24} < \frac{1}{e} < \frac{9}{24}$.
- For $n = 5$, we have $\frac{44}{120} < \frac{1}{e} < \frac{45}{120}$.
- For $n = 6$, we have $\frac{264}{720} < \frac{1}{e} < \frac{265}{720}$. And so on.

We infer that for every positive integer n , there exist consecutive positive integers C and $C + 1$ such that

$$\frac{C}{n!} < \frac{1}{e} < \frac{C+1}{n!},$$

i.e.,

$$C < \frac{n!}{e} < C + 1.$$

It follows that $n!/e$ never takes an integer value, for any positive integer n .

Hence $1/e$ is an irrational number; for, if $1/e = p/q$ for some two positive integers p, q , then q/e would be an integer, and therefore, $q!/e$ too would be an integer.

Since $1/e$ is irrational, so is e . ■

Remark

Since C depends on n , we should write C_n instead of C . Here are the values of C_n for small values of n . You may be able to spot a pattern!

n	3	4	5	6	7	8	9	10
C_n	2	8	44	264	1854	14832	133496	1334960

Further reading

[1] Sridhar Ramesh, "How do I prove that 'e' is an irrational number?"
from <https://tinyurl.com/2a65oh8u>

Area of an Octagon

Article contributed by A S Rajagopalan and Class 10 Maths Club students (of Rishi Valley School)

We consider here a geometric area problem that we solved during a session of the Maths Club in Rishi Valley School. We came across the problem in the book [1].

Problem. Find the area of a convex octagon inscribed in a circle, given that it has four sides of length 3 units and another four sides of length 2 units.

Note that we are not given the radius of the circumscribing circle.

What is of interest is that the problem permits many different approaches and strategies, illustrating different facets of geometry. We shall describe three of these approaches (the second one is from Larson's book).

First solution

The octagon may be thought of as the union of 8 isosceles triangles whose bases are the sides of the octagon. (The apex of each triangle is the centre of the circumscribing circle.) If we interchange any two of the triangles, the area of the octagon will not change. So we may consider the sides to be alternately 3, 2, 3, 2, 3, 2, 3, 2. Let the radius of the circle be r .

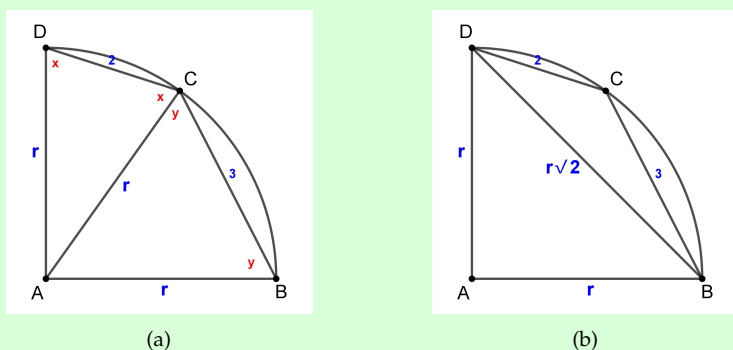


Figure 62

Consider the part $ABCD$ (Figure 62a) formed by the first two triangles. Note that the arc BCD is a quarter circle. We have now: $AB = AC = AD = r$, $BC = 3$, $CD = 2$. The angles marked x are equal, as are the angles marked y . Since $\angle DAB = 90^\circ$, it follows that $2(x + y) = 270^\circ$, so $x + y = 135^\circ$. Next, we compute the areas of triangles ABD and BCD . The area of $\triangle ABD$ is $\frac{1}{2}r^2$. The area of $\triangle BCD$ is

$$\frac{1}{2} \cdot 2 \cdot 3 \cdot \sin \angle BCD = 3 \sin 135^\circ = 3 \sin 45^\circ = \frac{3}{2}\sqrt{2}.$$

We find r by applying the cosine rule to $\triangle BCD$:

$$(r\sqrt{2})^2 = 2^2 + 3^2 - 2 \cdot 2 \cdot 3 \cdot \cos 135^\circ, \quad \therefore 2r^2 = 13 + 6\sqrt{2}.$$

The area of the octagon is 4 times the area of quadrilateral $ABCD$:

$$\begin{aligned} \text{Area of octagon} &= 4 \cdot \left(\frac{1}{2}r^2 + \frac{3}{2}\sqrt{2} \right) = 2r^2 + 6\sqrt{2} \\ &= 13 + 6\sqrt{2} + 6\sqrt{2} = 13 + 12\sqrt{2}. \end{aligned}$$

Hence the required area is $13 + 12\sqrt{2}$. ■

Second solution

This is surely the simplest solution; it is from [1] and is very elegant.

Let the octagon be $ABCDEFGH$ (Figure 63, below), with

$$AB = CD = EF = GH = 3, \quad BC = DE = FG = HA = 2.$$

We know from the earlier solution that each interior angle of the octagon is 135° . Hence each exterior angle of the octagon is 45° .

Draw lines through $\{B, C\}$, $\{D, E\}$, $\{F, G\}$ and $\{H, A\}$, as shown. These lines mark out a quadrilateral $PQRS$. Since the four triangles created outside the octagon are isosceles right-angled triangles (please justify this statement for yourself), $PQRS$ is a square. Denote the sides of these triangles by x . Then we have $x^2 + x^2 = 3^2$, hence $x^2 = 9/2$, $x = 3/\sqrt{2}$.

The area of the octagon is equal to the area of square $PQRS$ minus the area of the four triangles.

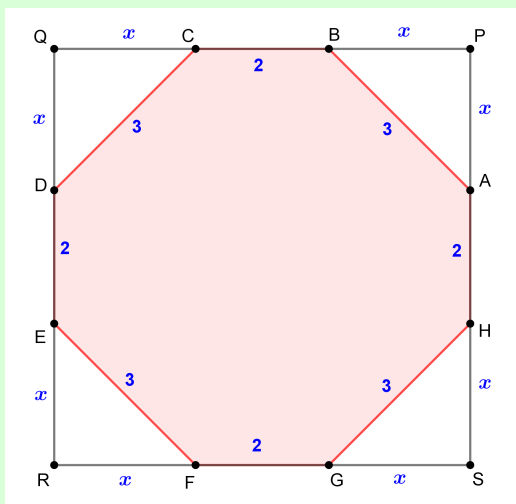


Figure 63

Hence:

$$\begin{aligned}
 \text{Area of octagon} &= (2x + 2)^2 - 4 \cdot \frac{x^2}{2} \\
 &= 4x^2 + 8x + 4 - 2x^2 = 2x^2 + 8x + 4 \\
 &= 9 + \frac{3 \cdot 8}{\sqrt{2}} + 4 = 13 + 12\sqrt{2}, \quad \text{as earlier.} \quad \blacksquare
 \end{aligned}$$

Third solution

We consider quadrilateral $ABCD$ as in the first solution; the area of the octagon is four times the area of the quadrilateral, as sector BAD is a quadrant of the circle.

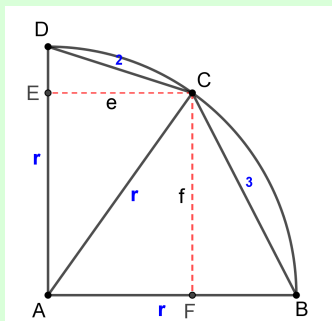


Figure 64

Drop perpendiculars $CE \perp AD$ and $CF \perp AB$ as shown (Figure 64). Let their lengths be e and f . Then $e^2 + f^2 = r^2$. Next,

$$\text{Area}(ABCD) = \text{Area}(\triangle ACD) + \text{Area}(\triangle ABC)$$

$$= \frac{1}{2}re + \frac{1}{2}rf = \frac{r(e+f)}{2},$$

$$\therefore \text{Area of octagon} = 2r(e+f).$$

Next, consider triangles CDE and CBF . We have:

$$CD^2 = DE^2 + CE^2$$

$$\therefore 2^2 = (r-f)^2 + e^2 = r^2 + e^2 + f^2 - 2rf = 2r^2 - 2rf,$$

and, similarly,

$$BC^2 = BF^2 + CF^2$$

$$\therefore 3^2 = (r-e)^2 + f^2 = r^2 + e^2 + f^2 - 2re = 2r^2 - 2re.$$

Hence by addition we get

$$\begin{aligned}2r^2 + 2r^2 - 2r(e + f) &= 13, \\ \therefore 2r(e + f) &= 4r^2 - 13.\end{aligned}$$

We find r as in the first solution (by applying the cosine rule to $\triangle BCD$); we get

$$2r^2 = 13 + 6\sqrt{2}.$$

Gathering together all the strands we get:

$$\begin{aligned}\text{Area of octagon} &= 2r(e + f) = 4r^2 - 13 \\ &= 26 + 12\sqrt{2} - 13 = 13 + 12\sqrt{2}.\end{aligned}$$

Further reading

- [1] Loren Larson, *Problem Solving Through Problems* (Springer-Verlag), Chapter 8, pages 280–285

About the author ...

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Triangle + Circle + Chord

Problem. Triangle ABC is right-angled at C (Figure 65). Its incircle touches BC at G . Line AD is drawn as shown so that $\angle ADC = 45^\circ$. Given that $AB = 5$, and $CG = DB$, find the length of chord EF .

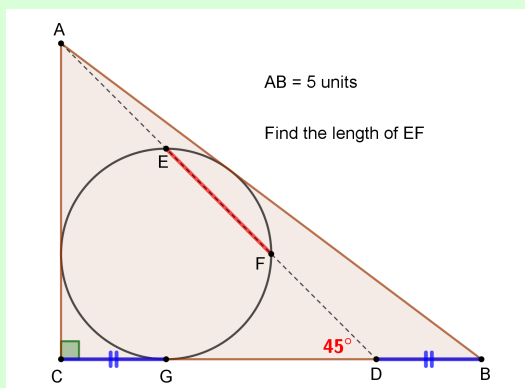


Figure 65

This is an attractive and challenging geometry problem. Please try it out yourself before reading on!

Solution of problem

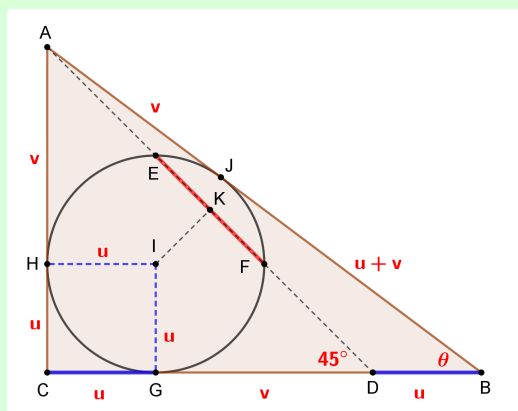


Figure 66

Assign symbols for the points (H, I, J) and lengths as shown: $CG = u = DB$, $GD = v$.

Then $CH = u$ and $AH = v$ (since $\angle ADC = 45^\circ$). Hence $AJ = v$ and $JB = u + v$ (remember that tangents to a circle from a point outside the circle have equal length). So $AB = u + 2v$.

Now invoke Pythagoras's theorem:

$$(u + 2v)^2 = (u + v)^2 + (2u + v)^2,$$

$$\therefore 2u^2 + uv - v^2 = 0,$$

$$\therefore (u + v)(2u - v) = 0,$$

so $v = 2u$ (since $u > 0, v > 0$). Hence $AJ = 2u$, $JB = 3u$, $AB = 5u$.

Since $AB = 5$, this yields $u = 1$ and thus $v = 2$.

Hence $AD = 3\sqrt{2}$ and $AK = 1.5\sqrt{2}$. (Here K is the midpoint of AD .)

Also, $AI^2 = 2^2 + 1^2 = 5$, so $AI = \sqrt{5}$.

Now consider $\triangle AIK$.

Since $AI = \sqrt{5}$ and $AK = 1.5\sqrt{2}$, we get $IK^2 = 5 - 4.5 = 0.5$, so $IK = \sqrt{0.5}$.

Hence $EK^2 = 1^2 - 0.5 = 0.5$, i.e., $EK = \sqrt{0.5}$.

Hence $EF = 2\sqrt{0.5} = \sqrt{2}$.

Answer: $x = \sqrt{2}$. ■

Comment 1

The answer obtained shows that $\triangle EIF$ is right-angled, with $\angle EIF = 90^\circ$. Could this have been anticipated in some other way? If yes, that would reveal another path to the answer.

Comment 2

The figure suggests that points E, I, G are collinear, and so are the points H, I, F . Is this so? Or is this an illusion?

Comment 3

The sides of $\triangle ABC$ are $AC = 3$, $BC = 4$, $AB = 5$. So the triangle in the figure is a 3-4-5 triangle!

Problems for Solution – 2

Here are some problems for you to ponder over. The answers are given on page 165.

Solutions to the problems in this set will be given in the next issue of the magazine. Enjoy!

Problems for the October 2024 issue

1. Prove that the fraction

$$\frac{21n + 4}{14n + 3}$$

is already in its “lowest terms” for every positive integer n . In other words, no common factor greater than 1 can be cancelled from the numerator and the denominator of the fraction.

In number theory such fractions are said to be *irreducible*; they cannot be ‘reduced’ further. Two examples of such fractions are $8/15$ and $6/25$.

Note: This problem was posed in the very first IMO in 1959.

2. Find all positive integers n such that $2^n + 1$ is a perfect square.
3. (a) Are there any positive integers n such that $n(n + 1)$ is a perfect square?
- (b) Are there any positive integers n such that $n(n + 1)(n + 2)$ is a perfect square?
4. Trapezium $ABCD$ has $AB \parallel DC$ and is such that the bisector of $\angle D$ is perpendicular to side BC (see Figure 67). The lengths of some of the segments are shown in the figure ($u = BF$, $v = CF$, $w = DF$). Find the area of trapezium $ABCD$ in terms of u, v, w .

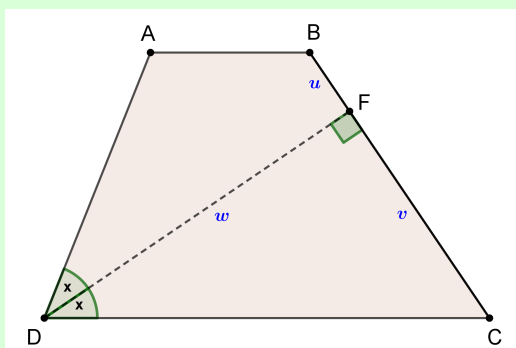


Figure 67

5. Find all positive integers a, b, c, d such that

$$a + \frac{1}{b + \frac{1}{c + \frac{1}{d}}} = \frac{421}{130}.$$

6. Find all the positive integers n such that $n^2 + 3^n$ is a perfect square.

Note: This problem was posed in the Fourth Junior Balkan Mathematical Olympiad, Macedonia, June 2000.

7. Find all fractions which can be written simultaneously in the forms

$$\frac{7a-5}{5k-3} \quad \text{and} \quad \frac{6b-1}{4b-3}$$

for some integers a, b .

Note: This problem was posed in the Indian Regional Mathematics Olympiad (RMO) in 2015.

Solutions to problems from the July 2024 issue

Problem 1 (July 2024): Are there any non-zero perfect squares that have two or more digits and are of the form $aaa \dots aa$? (Numbers are represented in base 10.)

Solution: We shall show that a number of the form $aaa \dots aa$ (with two or more a 's) cannot be a perfect square for any $a > 0$. We shall do so by eliminating all the available possibilities.

The units digit of a square must be one of the digits 0, 1, 4, 5, 6, 9. So we may restrict a to $\{1, 4, 5, 6, 9\}$.

We now make use of a simple result of great utility: *All square numbers are either divisible by 4 or leave remainder 1 under division by 4.* Otherwise expressed, all square numbers are of the form $4k$ or $4k + 1$.

Any number N can be expressed as $N = 100a + b$ where $0 \leq b \leq 99$. Since 100 is a multiple of 4, the remainder in the division $N \div 4$ is the same as

the remainder in the division $b \div 4$. (This is the basis for the well-known test for divisibility by 4.)

The numbers 11, 55 and 99 are of the form $4k + 3$; they leave remainder 3 under division by 4. It follows that there are no square numbers of the forms $\dots 11$ or $\dots 55$ or $\dots 99$ (where the dots represent arbitrary digits). Hence $a \neq 1, 5, 9$.

The number 66 is of the form $4k + 2$; it leaves remainder 2 under division by 4. It follows that there are no square numbers of the form $\dots 66$ (where the dots represent arbitrary digits). Hence $a \neq 6$.

The only option left is $a = 4$. If the number $N = 444\dots 44$ is a perfect square, then so is the number $N \div 4$ since 4 is a perfect square and it is a divisor of N . But $N \div 4 = 111\dots 11$, which is of the form $4k + 3$ and hence not a perfect square. It follows that $444\dots 44$ too is not a perfect square.

Therefore, there are no perfect squares of the form $aaa\dots aa$ (with two or more a 's) for any digit a .

Problem 2 (July 2024): Find the smallest perfect square whose digits start with 1234567. (That is, the number is of the form $1234567\dots$ where the dots represent more digits. Numbers are represented in base 10.)

Solution: Let $a^2 = 1234567\dots$ where a is a positive integer. We must differentiate between two possibilities here, depending on whether the number of digits in a^2 is even or odd. We consider both possibilities in turn.

Suppose the number of digits in a^2 is **even**. Then we can write:

$$a^2 = 1234567\dots = 12.34567\dots \times 10^{2n}$$

for some positive integer n . We clearly have

$$1234567000\dots \leq a^2 < 1234568000\dots$$

Hence

$$12.34567 \times 10^{2n} \leq a^2 < 12.34568 \times 10^{2n},$$

and so

$$\sqrt{12.34567} \times 10^n \leq a < \sqrt{12.34568} \times 10^n.$$

On computing the two square roots, we get:

$$3.513640562 \times 10^n \leq a < 3.513641985 \times 10^n.$$

As we have not specified the value of n , this may equally well be written as:

$$3513640.562 \times 10^{n'} \leq a < 3513641.985 \times 10^{n'}$$

for some positive integer n' . On looking closely at the numbers, we realize that the least value of a that fits the requirement is 3513641. And indeed we have:

$$3513641^2 = 12\,345\,673\,076\,881.$$

This is the smallest perfect square with an even number of digits that starts 1234567.... Observe that it has 14 digits.

Now suppose the number of digits in a^2 is **odd**. Then we can write:

$$a^2 = 1234567\dots = 1.234567\dots \times 10^{2n}$$

for some positive integer n . We clearly have

$$1234567000\dots \leq a^2 < 1234568000\dots$$

Hence

$$1.234567 \times 10^{2n} \leq a^2 < 1.234568 \times 10^{2n},$$

and so

$$\sqrt{1.234567} \times 10^n \leq a < \sqrt{1.234568} \times 10^n.$$

On computing the two square roots, we get:

$$1.111110705 \times 10^n \leq a < 1.111111155 \times 10^n.$$

As we have not specified the value of n , this may equally well be written as:

$$1111110.705 \times 10^{n'} \leq a < 1111111.155 \times 10^{n'}$$

for some positive integer n' . On looking closely at the numbers, we realize that the least value of a that fits the requirement is 1111111. And indeed we have

$$1111111^2 = 1\,234\,567\,654\,321.$$

This is the smallest perfect square with an odd number of digits that starts 1234567.... Observe that it has 13 digits.

As this has fewer digits than what we got the previous time, this is the required answer. So the smallest perfect square that starts 1234567... is 1 234 567 654 321.

Problem 3 (July 2024): How many distinct integers n are there (positive or negative) such that $n^2 + 20n + 24$ is a perfect square?

Solution: Let $n^2 + 20n + 24 = m^2$ where m is an integer. We shall make use of the well-known method for “completing the square.” We have:

$$n^2 + 20n + 24 = m^2,$$

$$\therefore (n^2 + 20n + 100) - 76 = m^2,$$

$$\therefore (n + 10)^2 - m^2 = 76,$$

$$\therefore (n + 10 + m) \cdot (n + 10 - m) = 76.$$

We see that the product of the two integers $n + 10 + m$ and $n + 10 - m$ is 76. Hence, to find all possible values of n , we must consider all possible ways of expressing 76 as a product of two integer factors. We can discard factorizations in which the two factors have opposite parity; for, $n + 10 + m$ and $n + 10 - m$ necessarily have the same parity (because their sum is even). The only such factorizations are

$$2 \times 38, \quad 38 \times 2, \quad (-2) \times (-38), \quad (-38) \times (-2).$$

By addition it follows that $2n + 20 = 40$ or -40 , giving $n = 10$ or -30 . So there are just two integers n such that $n^2 + 20n + 24$ is a perfect square; namely, 10 and -30 . Let $f(n) = n^2 + 20n + 24$. We may check that

$$f(10) = 324 = 18^2,$$

$$f(-30) = 324 = 18^2.$$

It is curious that both values yield the same perfect square (18^2). Why?

Problem 4 (July 2024): Which is larger,

$$1.0001 + \frac{1}{1.0001} \quad \text{or} \quad 1.001 + \frac{1}{1.001}?$$

Solution: The given expressions are both of the kind $x + 1/x$ for values of x close to 1. Now observe that for $x > 0$,

$$\left(x + \frac{1}{x}\right)^2 = \left(x - \frac{1}{x}\right)^2 + 4.$$

From this we can conclude that for $x > 1$, as the gap between x and $1/x$ widens, the value of $x + 1/x$ increases.

In particular, we can conclude that if $a > b > 1$, then $a - 1/a > b - 1/b$, and therefore, $a + 1/a > b + 1/b$. Therefore:

$$1.001 + \frac{1}{1.001} > 1.0001 + \frac{1}{1.0001}.$$

In case you are still sceptical, here are the actual values! —

$$1.001 + \frac{1}{1.001} = 2.000\ 000\ 999\ 000\ 99\dots$$

$$1.0001 + \frac{1}{1.0001} = 2.000\ 000\ 009\ 999\ 00\dots$$

Convinced?!

Here is another way of working out the answer (which you may prefer).

Let $a > b > 1$; then:

$$\begin{aligned} \left(a + \frac{1}{a}\right) - \left(b + \frac{1}{b}\right) &= (a - b) + \left(\frac{1}{a} - \frac{1}{b}\right) \\ &= (a - b) + \frac{b - a}{ab} \\ &= (a - b) \left(1 - \frac{1}{ab}\right) > 0, \end{aligned}$$

since $a - b > 0$ and $ab > 1$ (so $1/ab < 1$). Hence $a + 1/a > b + 1/b$.

Problem 5 (July 2024): Which is larger, $200!$ or 100^{200} ?

Solution: Here too, let us proceed as though we do not have any computing facilities. In such problems, it is good to experiment with smaller numbers so as to build up one's intuitive sense of "what is happening." So let us compare $(2n)!$ and n^{2n} for a few (small) values of n :

- Let $n = 2$. Then $(2n)! = 4! = 24$, and $n^{2n} = 2^4 = 16$.
- Let $n = 3$. Then $(2n)! = 6! = 720$, and $n^{2n} = 3^6 = 729$.
- Let $n = 4$. Then $(2n)! = 8! = 40320$, and $n^{2n} = 4^8 = 65536$.
- Let $n = 6$. Then $(2n)! = 12! = 479001600$, and $n^{2n} = 6^{12} = 2176782336$.
- Let $n = 10$. Then $(2n)! = 20! = 2432902008176640000 \approx 2.43 \times 10^{18}$, and $n^{2n} = 10^{20}$.

The results are most revealing!

Apart from the initial value $n = 2$ (which now begins to look like an outlier — to use a term more commonly used in data analysis), in each case it is n^{2n} which is the larger number.

So we conjecture on the basis of these results that

$$n^{2n} > (2n)! \quad \text{for all } n > 2.$$

This is indeed true and we shall prove this using a well-known result: the arithmetic mean-geometric mean (AM-GM) inequality. This is the statement that for any two unequal positive numbers, their arithmetic mean (AM) exceeds their geometric mean (GM). That is if a, b are positive numbers with $a \neq b$, then

$$\frac{a+b}{2} > \sqrt{ab}.$$

We may also state this as: if $a, b > 0$ and $a \neq b$, then

$$\left(\frac{a+b}{2}\right)^2 < ab.$$

We apply this result to our problem as follows. We have:

$$\left(\frac{1+(2n-1)}{2}\right)^2 > 1 \times (2n-1)$$

$$\left(\frac{2+(2n-2)}{2}\right)^2 > 2 \times (2n-2)$$

$$\dots \dots > \dots \dots$$

$$\left(\frac{(n-1)+(n+1)}{2}\right)^2 > (n-1) \times (n+1).$$

We also have :

$$n^2 > 2n \quad (\text{since } n > 2).$$

Multiplying the corresponding sides of all these inequalities we obtain:

$$(n)^{2n} > (2n)! \quad \text{for all } n > 2.$$

This is just what we had set out to prove. In consequence we have:

$$100^{200} > 200!$$

The reader might want to see how these two numbers stand in relation to each other. We have:

$$100^{200} = 10^{400}, \quad 200! \approx 7.89 \times 10^{374}.$$

There is a substantial difference between the two numbers!

Answers to “Puzzles to Puzzle You – 2”

We give here the answers to the cryptarithms on page 29. Note that these are only the answers; we will give the solutions in the next issue.

1. We shall give the answers in the next issue (for all numbers from 1 to 100).
2. There are five dogs, four cats, two ducks, and three rabbits.
3. There is no missing ₹10. The puzzle has been deliberately written in a way that confuses the reader. The three friends spend ₹270. Of this, ₹250 is for the meal, and ₹20 is for the tip. The ₹30 they get back is entirely irrelevant! So the relevant calculation is $270 - 20 = 250$, and **not** $270 + 20 = 290$.
4. $138 \times 138 = 19044$

Answers to “Problems for Solution” – 2

We give here the answers to the problems on page 154.

Note that these are only the answers; we will give the solutions in the next issue.

2. $n = 3$ is the only solution, with $2^3 + 1 = 3^2$.
3. No such positive integers in either case.
5. The only solution in positive integers is $a = 3, b = 4, c = 5, d = 6$.
6. The positive integers n satisfying the stated condition are $n = 1$ and $n = 3$.
7. The fractions that can be written simultaneously in the two forms are the following:

$$\frac{5}{3}, 1, \frac{37}{27}, \frac{43}{31}, \frac{19}{13}, \frac{13}{9}, \frac{61}{43}, \frac{30}{19}.$$

Call for Articles

We invite mathematics teachers and practitioners to contribute short articles to The Compass.

Articles could deal with mathematical content or the teaching of mathematics (classes 6 to 12). They could feature real-life classroom experiences with projects and explorations.

They could be about the use of audiovisual material or social media in the mathematics classroom.

They could feature reflections on the teaching of mathematics, or on the nature of mathematics.

Articles about the use of software are particularly welcome, as are articles dealing with non-routine problems and classroom experiences with problem-solving.

In short, anything at all to do with school mathematics!